Course Notes

Lecture 08 — IP Networks for IoT: IPv4, NAT, IPv6 and Neighbor Discovery

Communication Technologies for the Internet of Things

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Text to support independent study

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How to Use These Notes

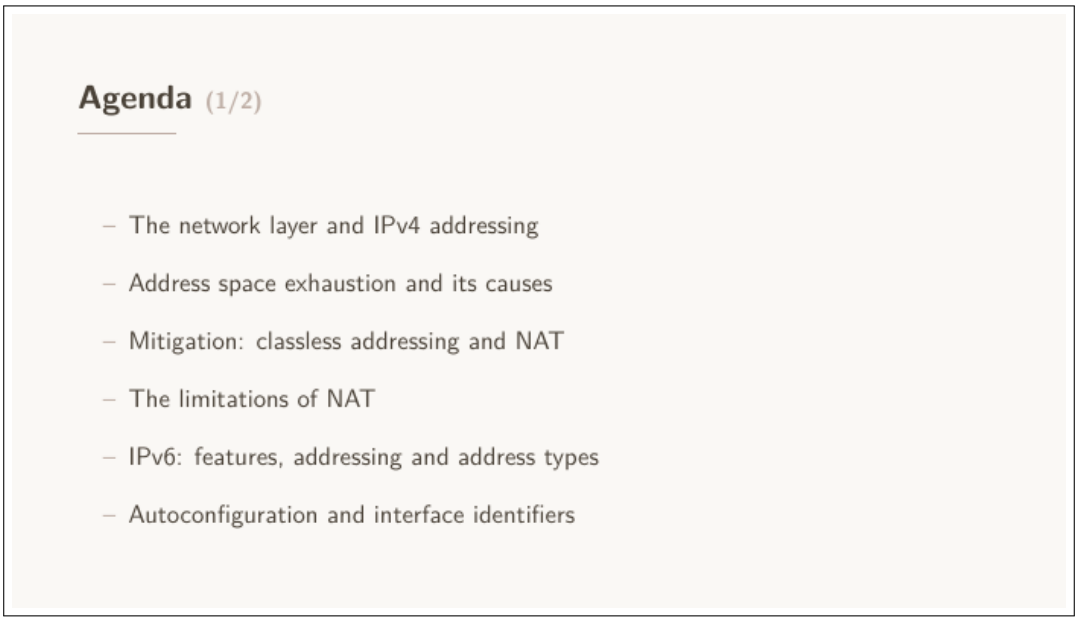
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

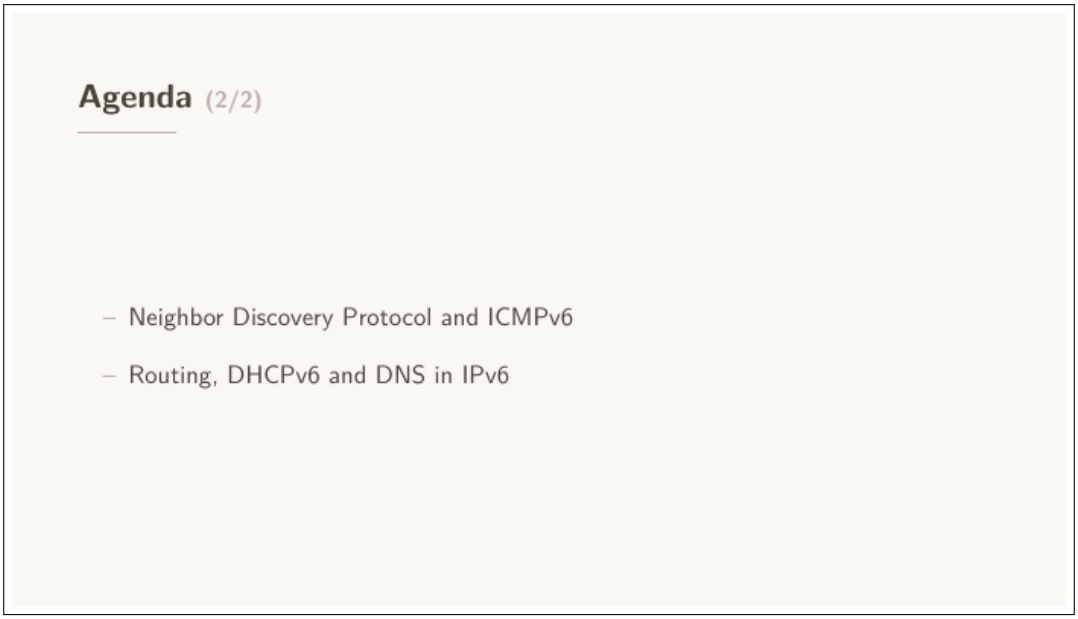
1. Commentary on the Slides

Slide 2. Agenda (1/2)



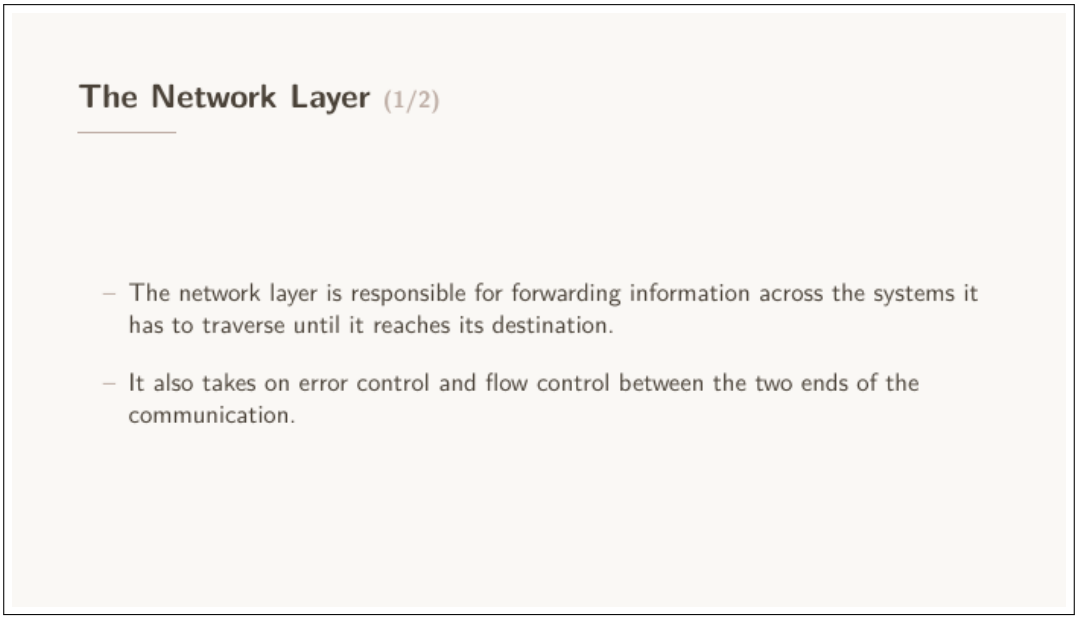
The path through the lecture. The lecture starts with what IPv4 is and why it ran out, moves on to the solutions that postponed the problem, and ends with the structural solution, IPv6, with everything that comes with it: address types, autoconfiguration and neighbour discovery. The sequence tells a story of cause and consequence, not a catalogue of protocols.

Slide 3. Agenda (2/2)



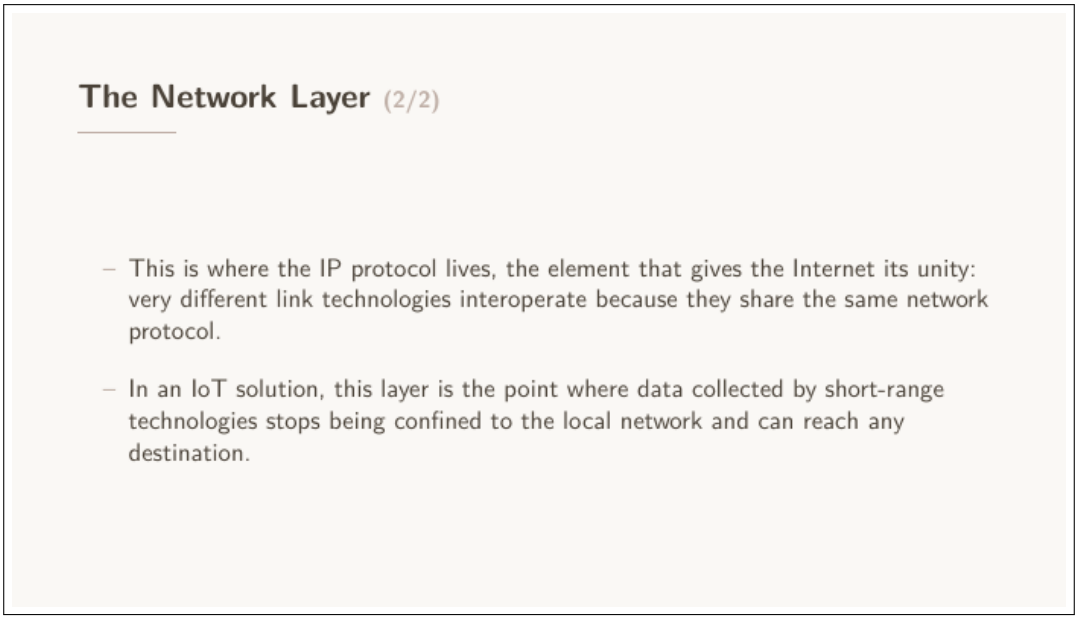
The second half of the agenda. Neighbor Discovery and ICMPv6 are what make IPv6 usable without manual configuration, and routing, DHCPv6 and DNS complete the picture of an operational network. For the Internet of Things, autoconfiguration is where the lecture comes closest to the real problem of constrained devices.

Slide 4. The Network Layer (1/2)



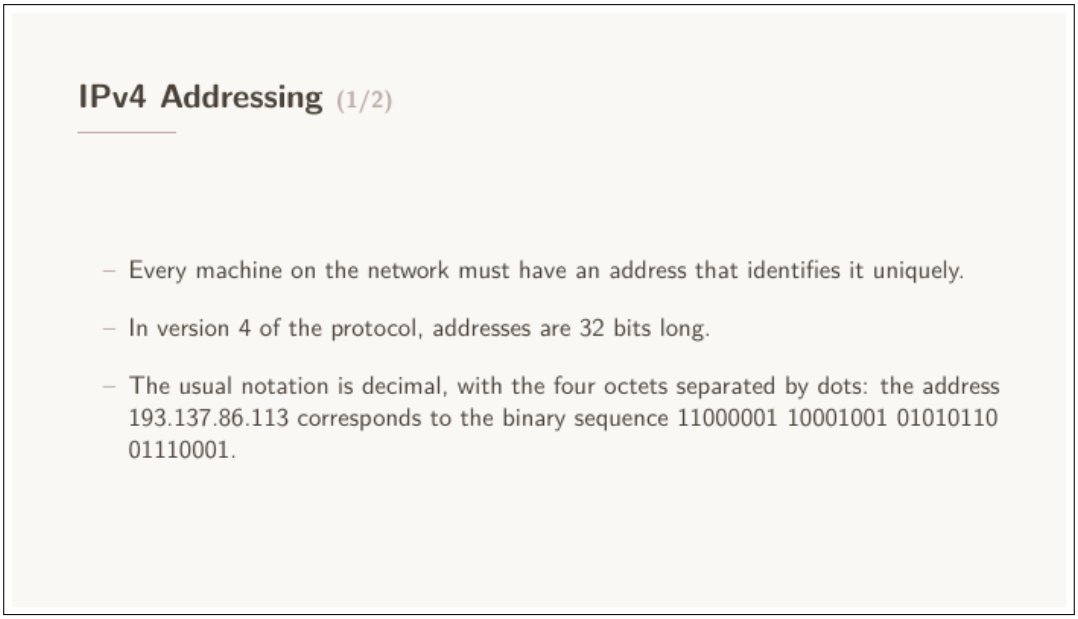
The role of the network layer. The network layer solves the problem of carrying a packet from one end to the other, across intermediate systems that know neither the source nor the destination. Forwarding is its central function; error and flow control between the ends, mentioned on the slide, is a function that the model also assigns, in practice, to higher layers.

Slide 5. The Network Layer (2/2)



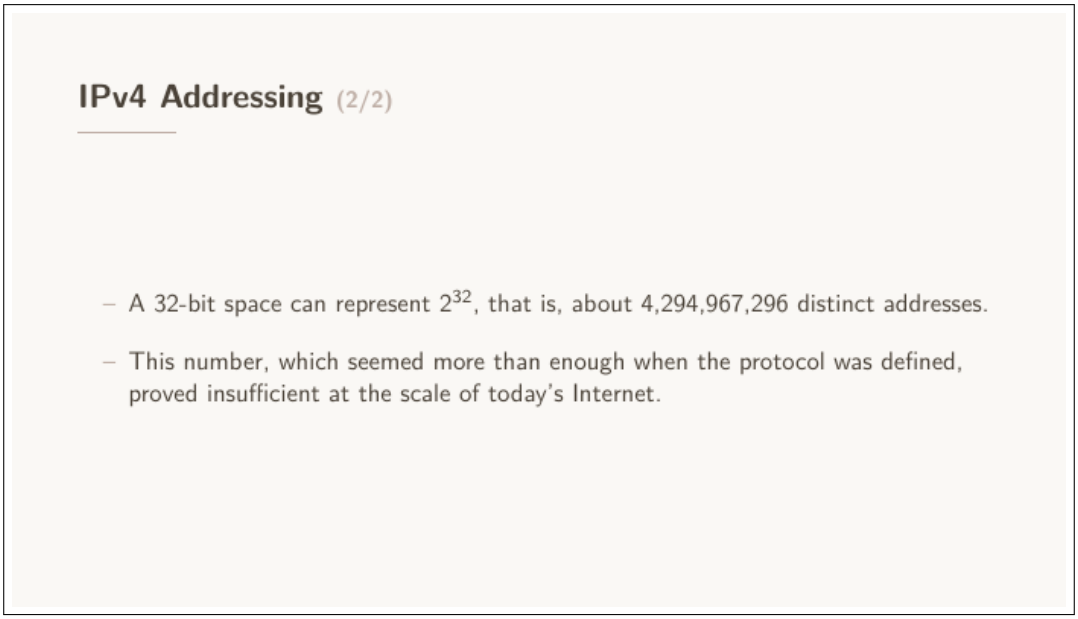
What IP unifies. IP is the narrow waist of the Internet: below it there are very different technologies, above it very different applications, and all of them meet in a single packet format. In IoT this is decisive: a short-range technology only reaches the cloud because, at some point, its data starts travelling in IP packets.

Slide 6. IPv4 Addressing (1/2)



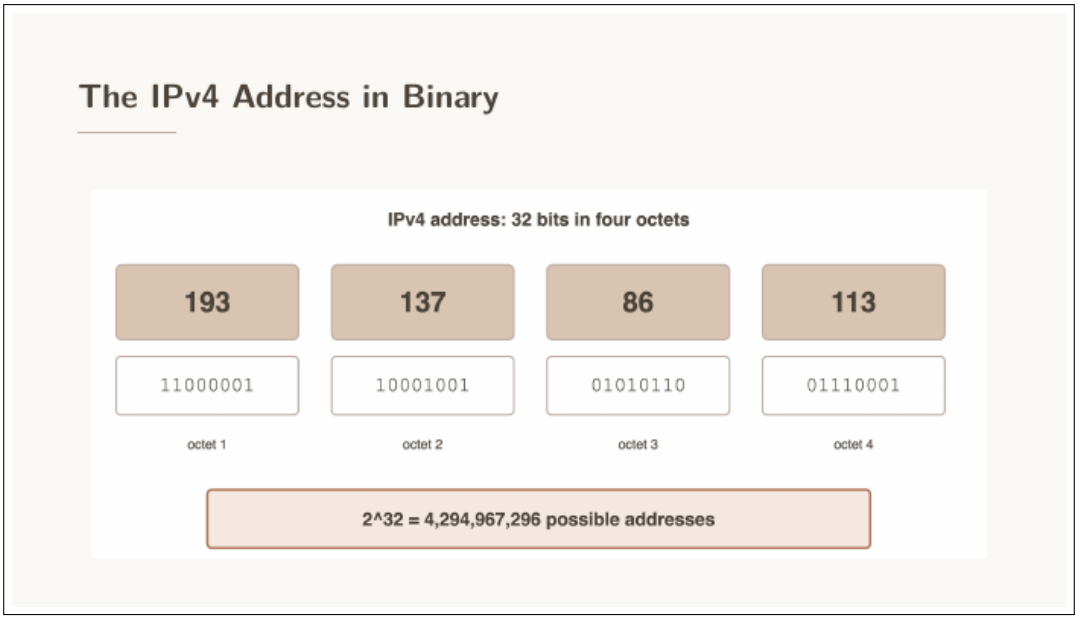
One address, one machine. An IPv4 address has 32 bits and identifies an interface on the network. Dotted-decimal notation is only a convenience for humans: what forwarding uses is the binary sequence. You should be able to convert an octet from decimal to binary, because classless addressing and subnet masks only make sense in binary.

Slide 7. IPv4 Addressing (2/2)



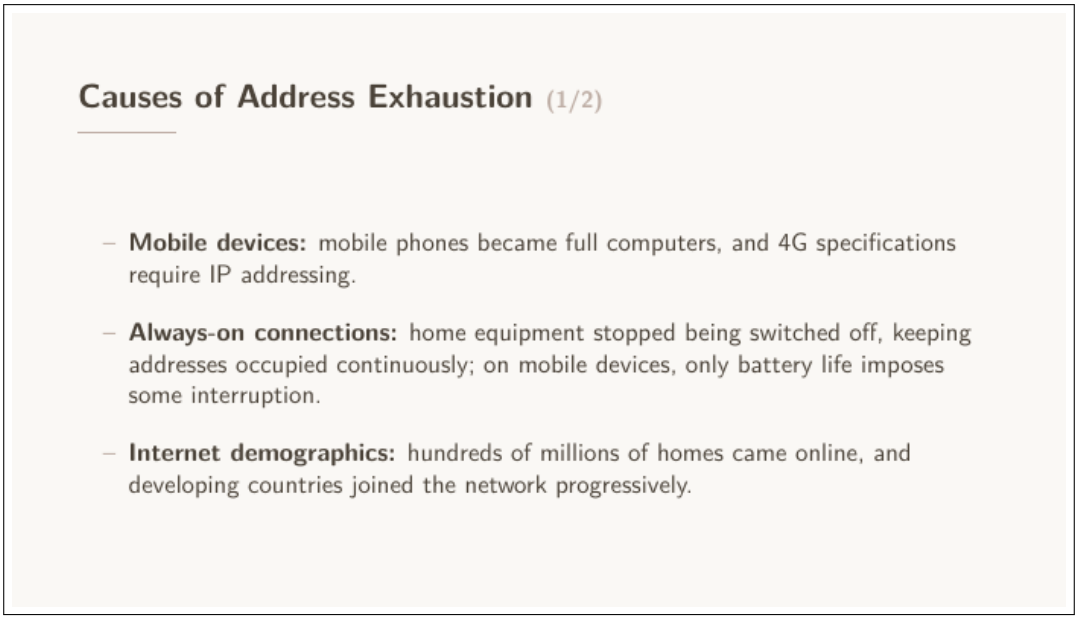
Why 32 bits stopped being enough. Two to the power of 32 gives a little over four billion addresses, a number that seemed enormous when the Internet was an academic network. Today there are more connected devices than possible addresses, and a considerable part of the space is allocated inefficiently.

Slide 8. The IPv4 Address in Binary



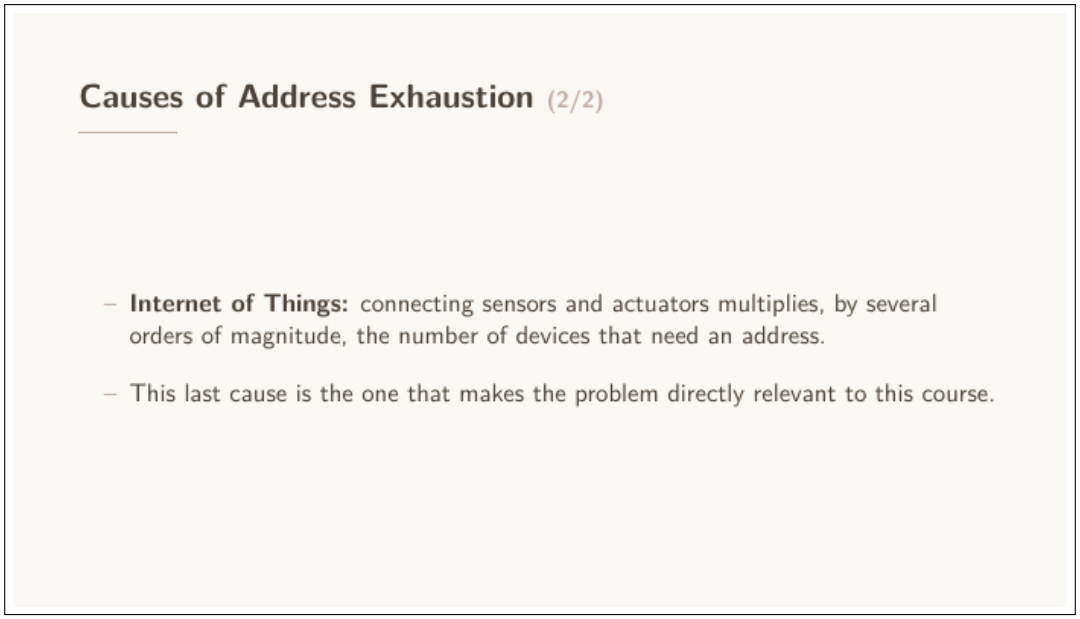
Reading the address in binary. The diagram shows the four octets of the address 193.137.86.113 and their binary representation. Each octet runs from 0 to 255 because it has 8 bits. Check: can you convert 86 to binary without looking at the slide?

Slide 9. Causes of Address Exhaustion (1/2)



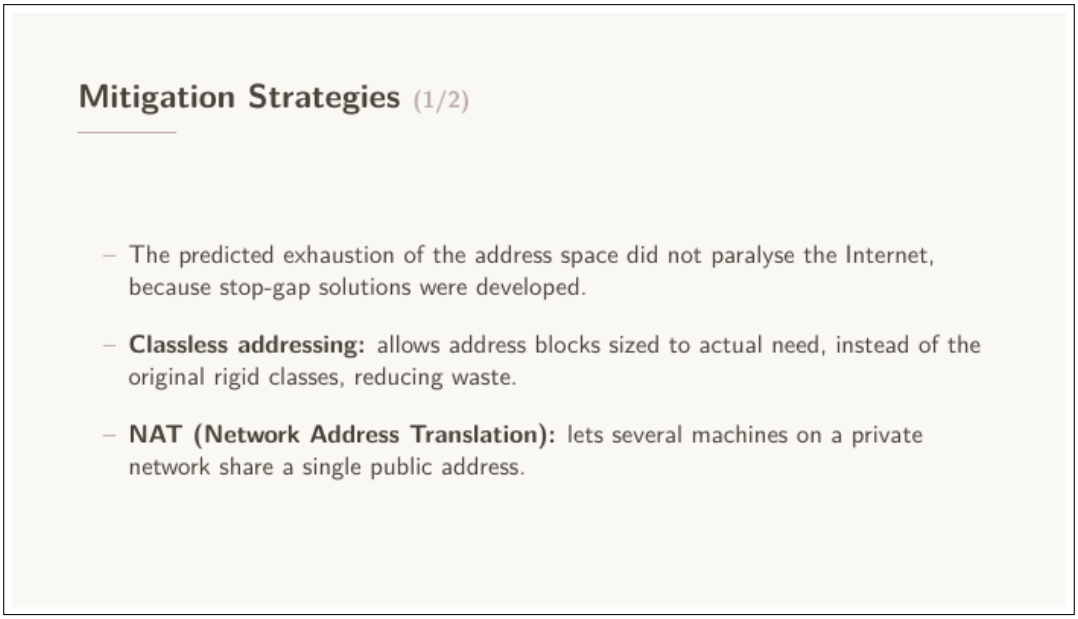
Three causes that predate IoT. Mobile devices, always-on connections and the arrival of new countries on the Internet explain why exhaustion was already visible before IoT existed as a market. In each case, what grows is the number of machines that need an address at the same time.

Slide 10. Causes of Address Exhaustion (2/2)



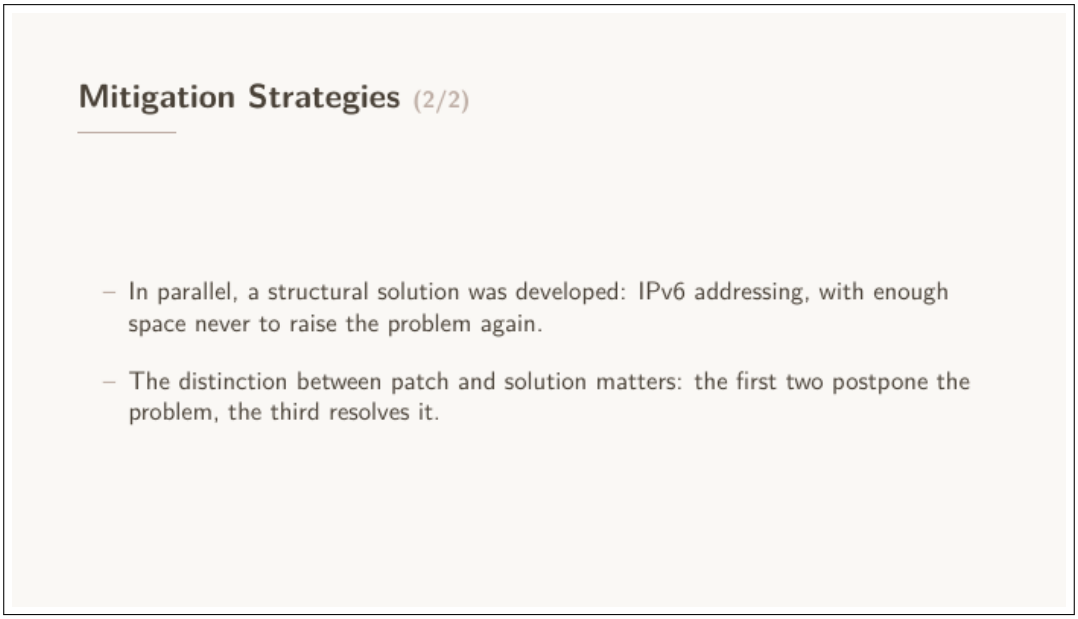
The cause that matters here. The Internet of Things is the cause that changes the order of magnitude. A patient, a building or a farm now has dozens or hundreds of sensors, each needing to communicate. This is why the course treats addressing as a topic in its own right.

Slide 11. Mitigation Strategies (1/2)



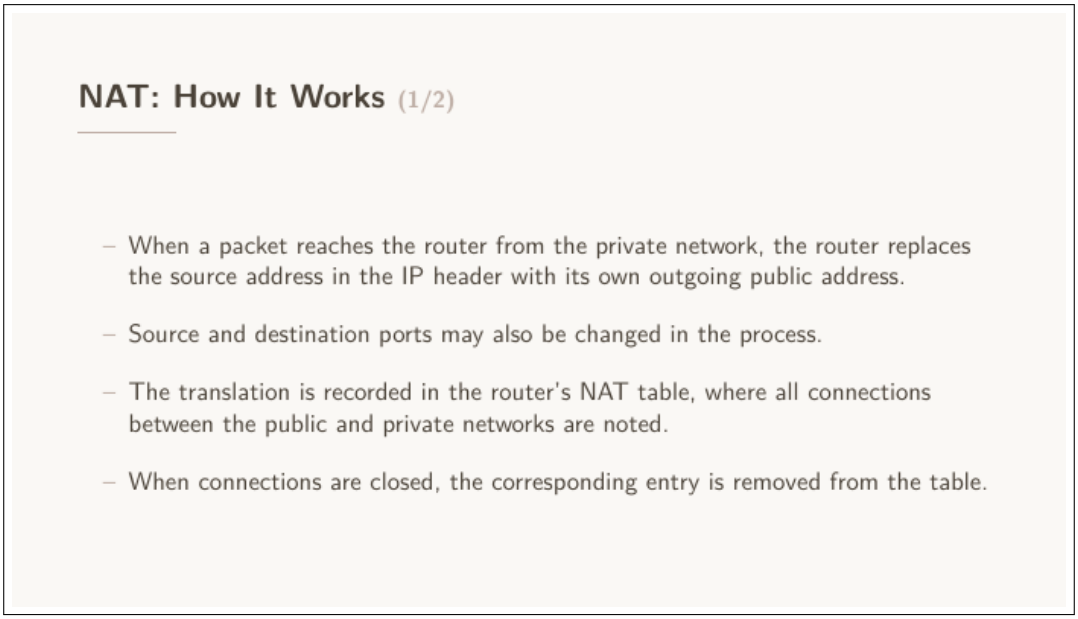
Patches that worked. Classless addressing and NAT were pragmatic responses. The first limits waste, because blocks are now the size that is needed. The second allows the same public address to be reused by many private machines. Neither increases the number of public addresses available.

Slide 12. Mitigation Strategies (2/2)



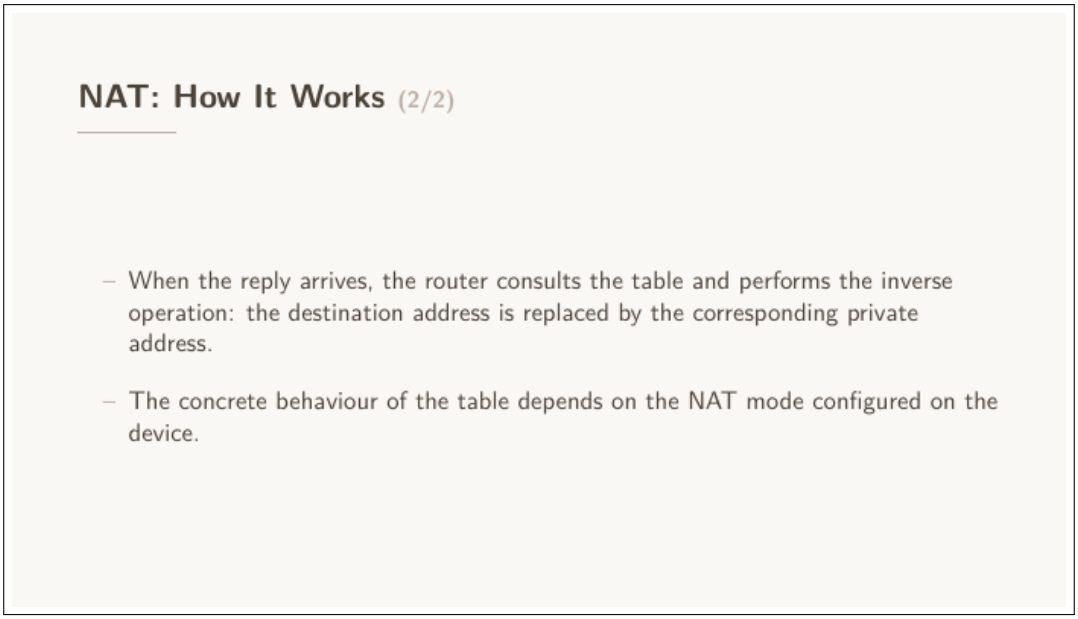
Patch versus solution. The distinction on the slide is the idea to retain: postponing a problem is not solving it. IPv6 is the only one of the three responses that changes the cause, which is the size of the address space.

Slide 13. NAT: How It Works (1/2)



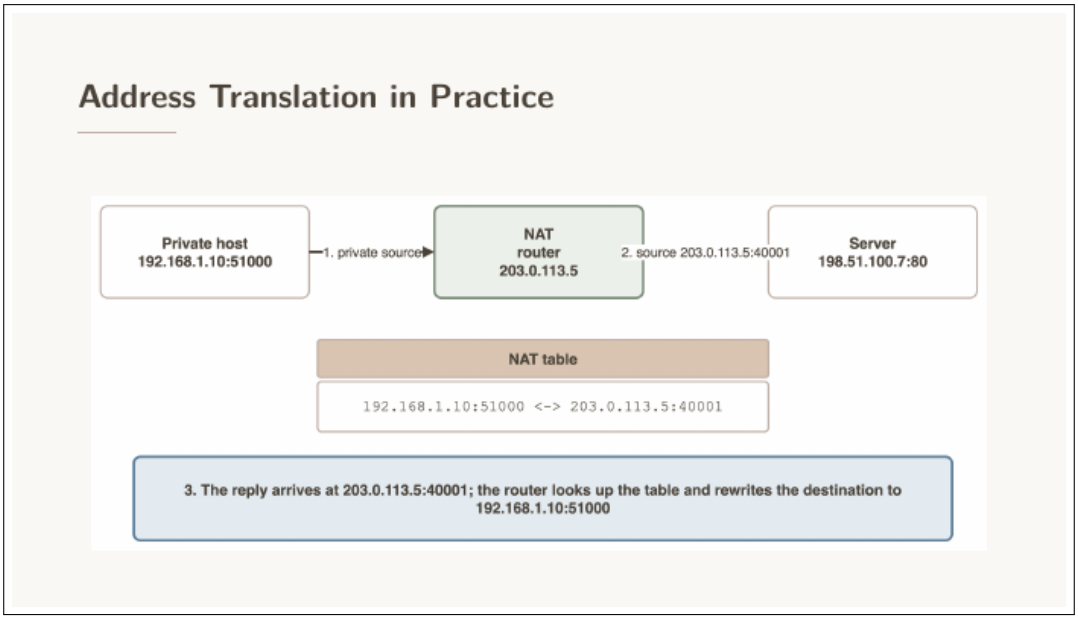
Translation step by step. The border router rewrites the source address of outgoing packets and records that rewrite in a table. It is the table that makes the inverse operation possible. Without it, the reply would reach the router's public address and the router would not know which internal machine to hand it to.

Slide 14. NAT: How It Works (2/2)



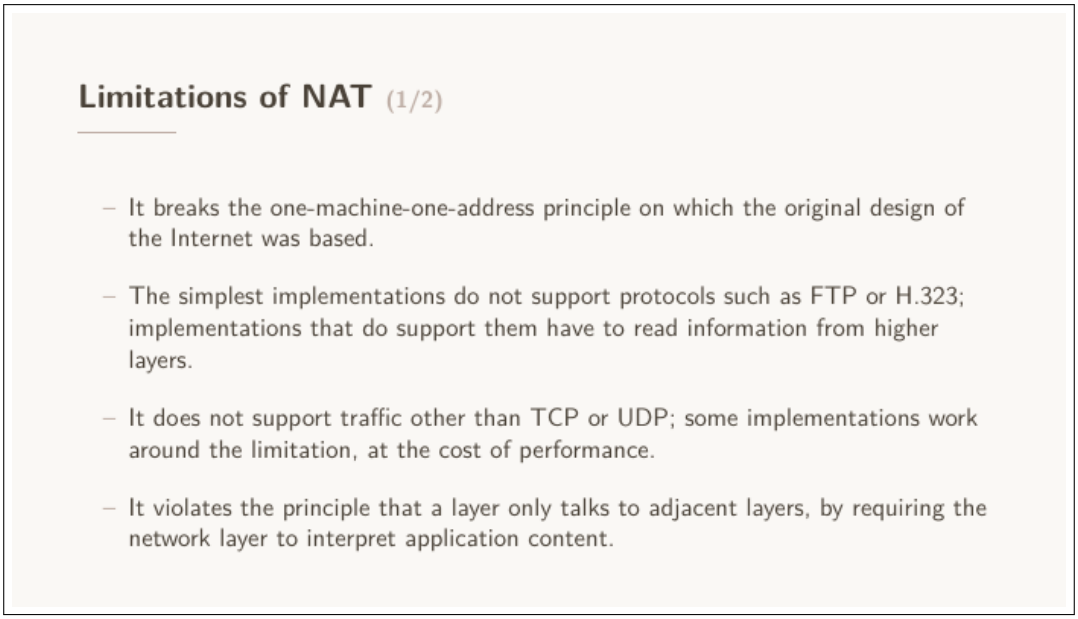
The inverse operation. On the reply, the router consults the table and restores the private address. Two practical details: the entry exists only while the connection exists, and the configured NAT mode determines what happens to unsolicited traffic.

Slide 15. Address Translation in Practice



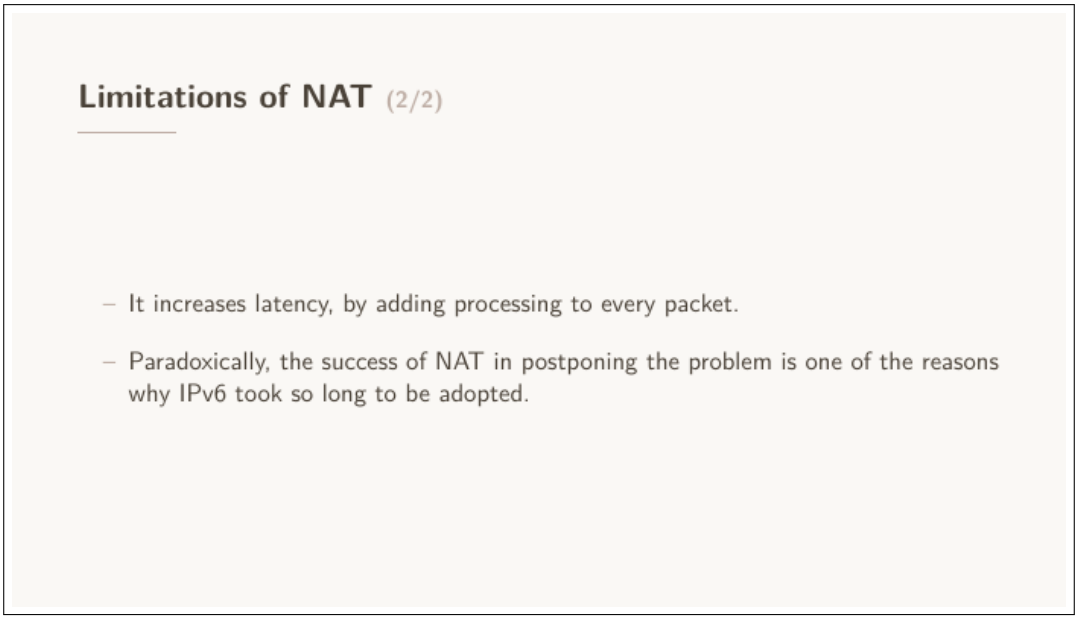
NAT in an example. The figure shows the same connection seen from three positions. The private host uses 192.168.1.10 and port 51000. The server sees only 203.0.113.5 and port 40001. The NAT table, in the centre, is the only element that knows both versions. Note that the port was also changed: that is what allows several private hosts to share the same public address.

Slide 16. Limitations of NAT (1/2)



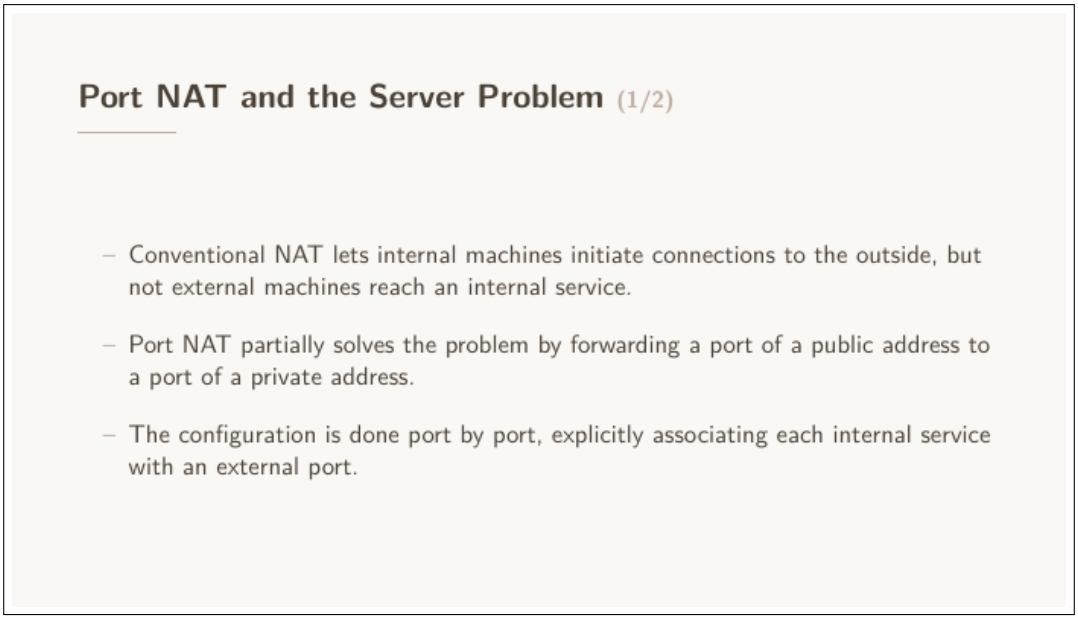
What NAT breaks. NAT violates the end-to-end model, in which two systems communicate directly with each other. Protocols that carry addresses in their own content, such as FTP, stop working without special handling, because the address they carry is no longer the one the network sees.

Slide 17. Limitations of NAT (2/2)



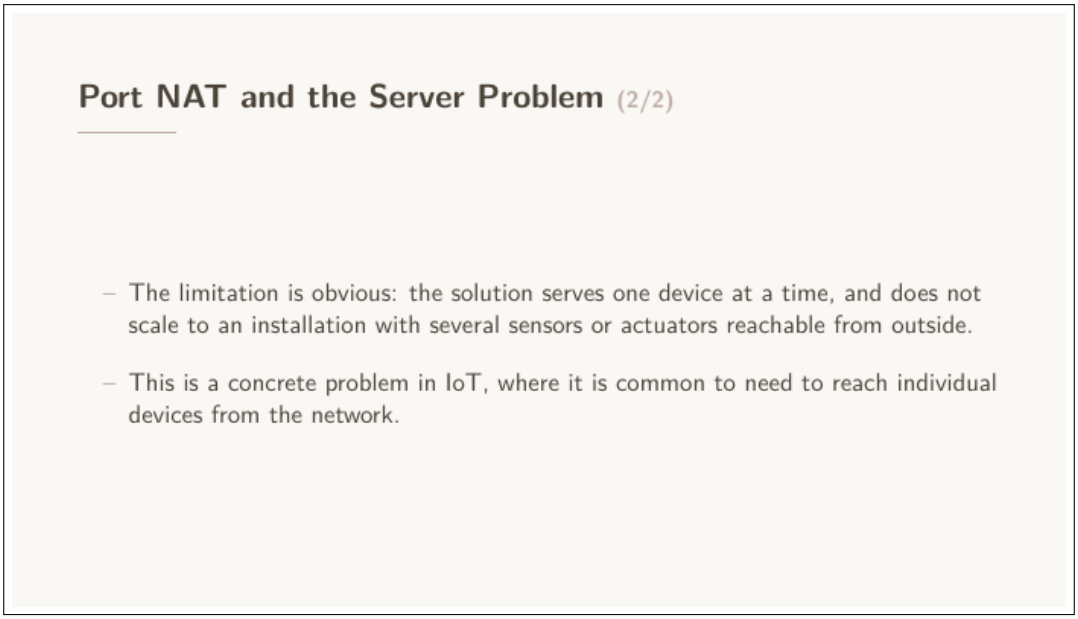
Latency and inertia. NAT adds processing to every packet. The subtlest point is the last one: by working well enough, it delayed the motivation to migrate to IPv6. What solved the immediate problem prolonged the structural one.

Slide 18. Port NAT and the Server Problem (1/2)



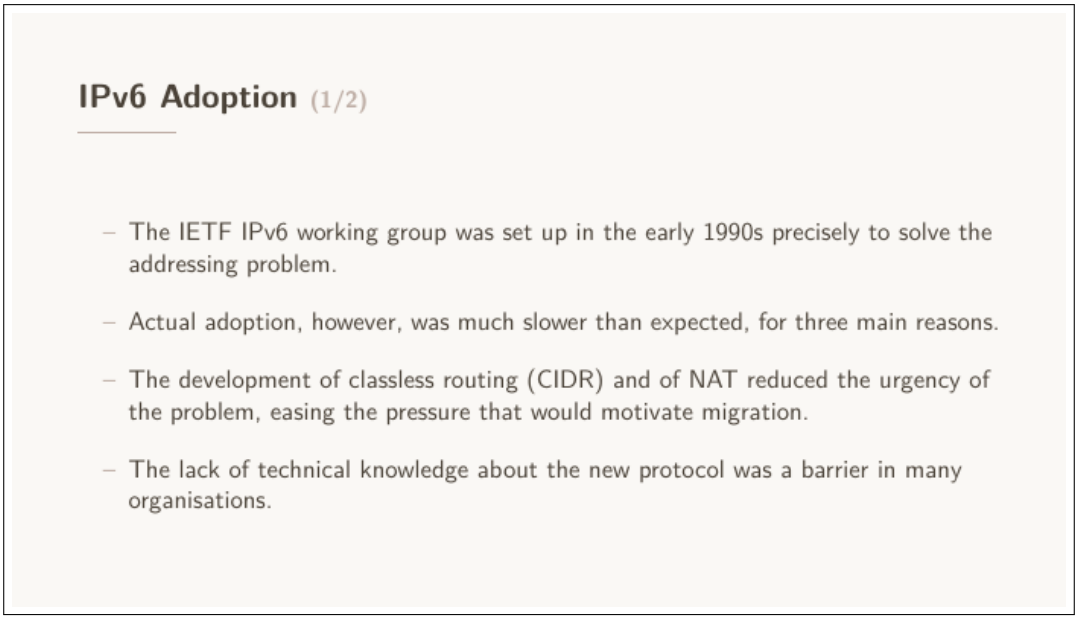
Servers behind NAT. An internal machine can initiate connections to the outside, but nobody outside can initiate a connection to it, because there is no entry in the NAT table. Port NAT creates that entry by hand, associating a public port with a private service.

Slide 19. Port NAT and the Server Problem (2/2)



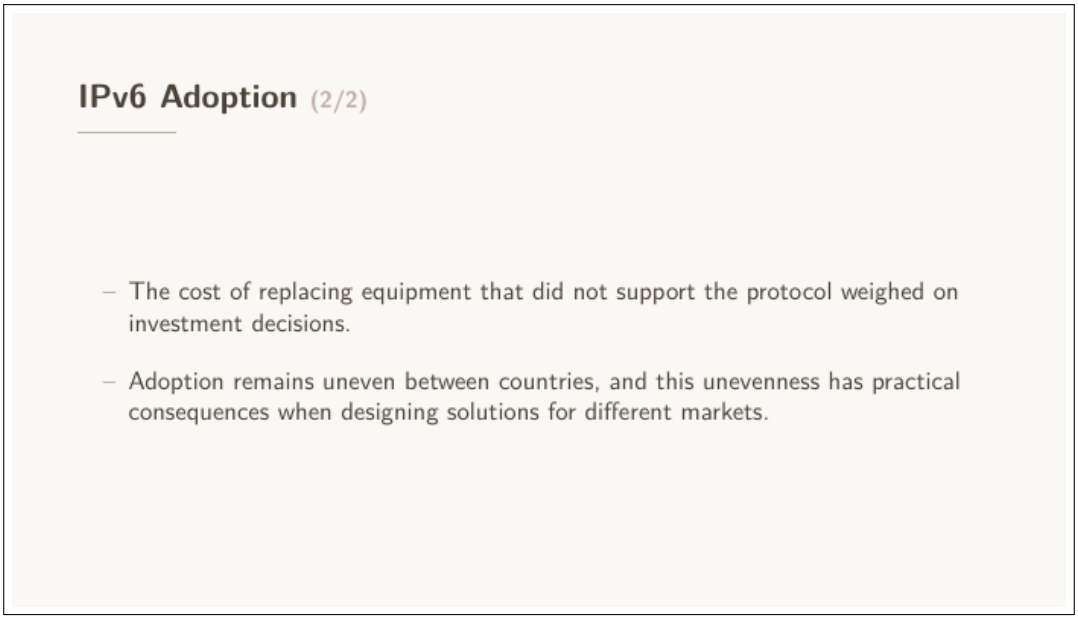
The limit of Port NAT. Associating a port with a service serves one machine at a time and requires manual configuration. In an installation with dozens of sensors reachable from outside, that is not manageable. It is a concrete IoT problem, not a networking curiosity.

Slide 20. IPv6 Adoption (1/2)



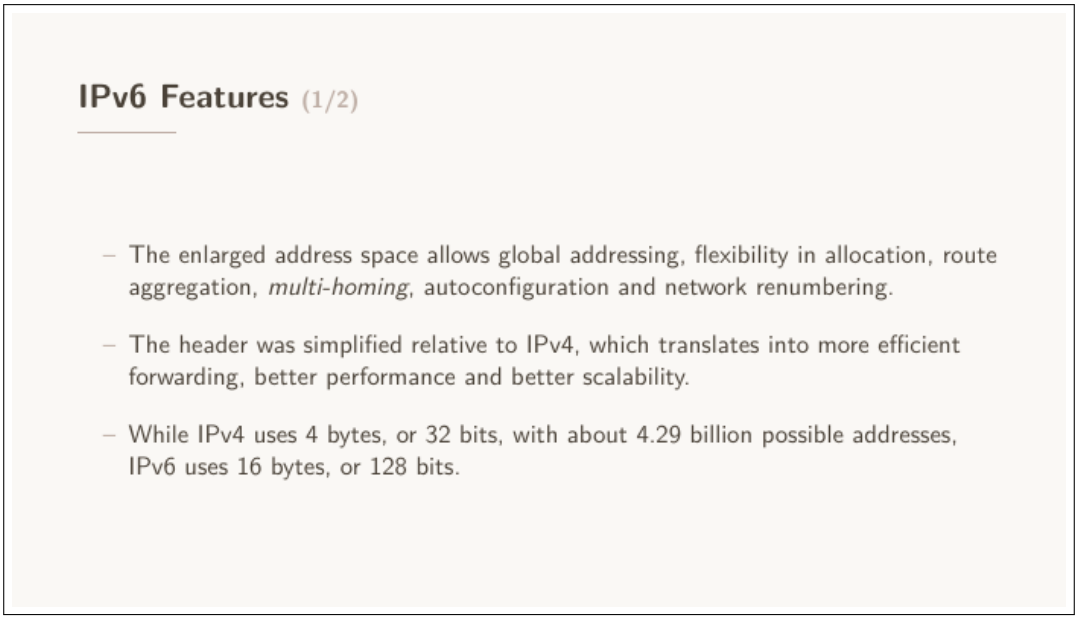
A thirty-year-old solution. IPv6 is not new: the IETF working group was created in the early 1990s. The slow adoption was not due to flaws in the protocol, but to the fact that CIDR and NAT eased the pressure and that migration has real costs.

Slide 21. IPv6 Adoption (2/2)



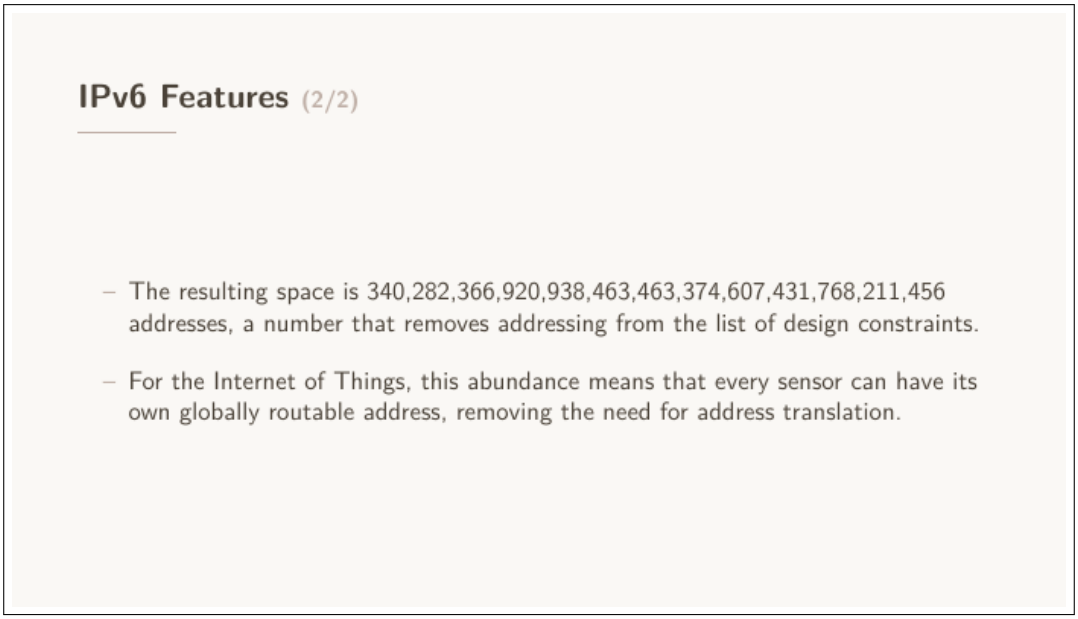
Uneven adoption. The cost of replacing equipment and the unevenness between countries have a practical consequence for anyone designing solutions: a product aimed at several markets has to work on IPv4 networks, on IPv6 networks and on networks with both.

Slide 22. IPv6 Features (1/2)



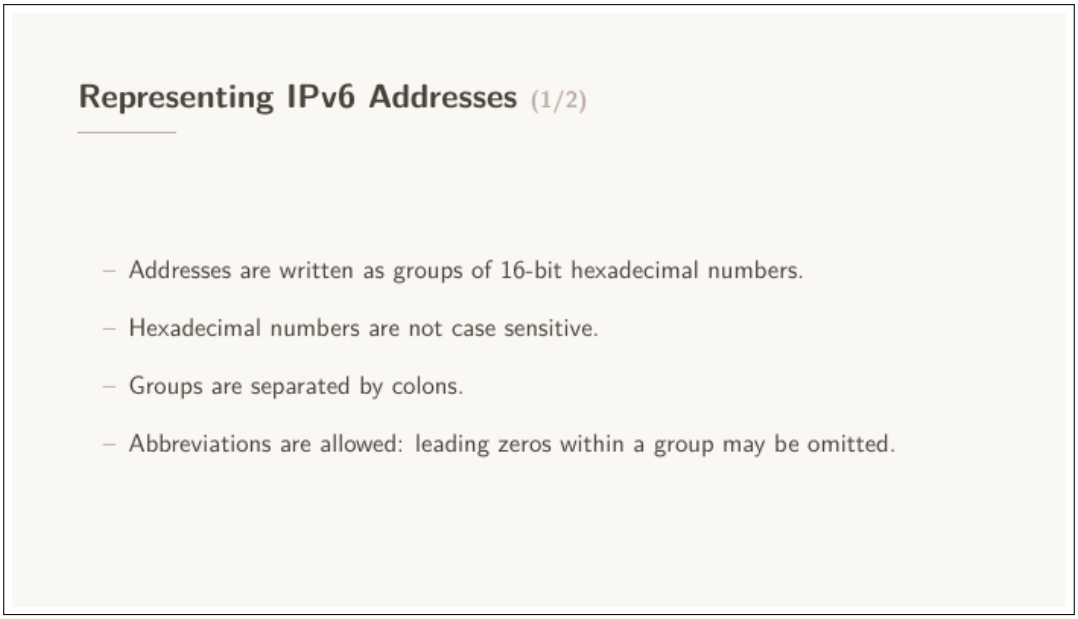
More than more addresses. IPv6 is not just a bigger address. The header was simplified, which reduces the work of routers, and features that were add-ons in IPv4, such as autoconfiguration, became part of the protocol.

Slide 23. IPv6 Features (2/2)



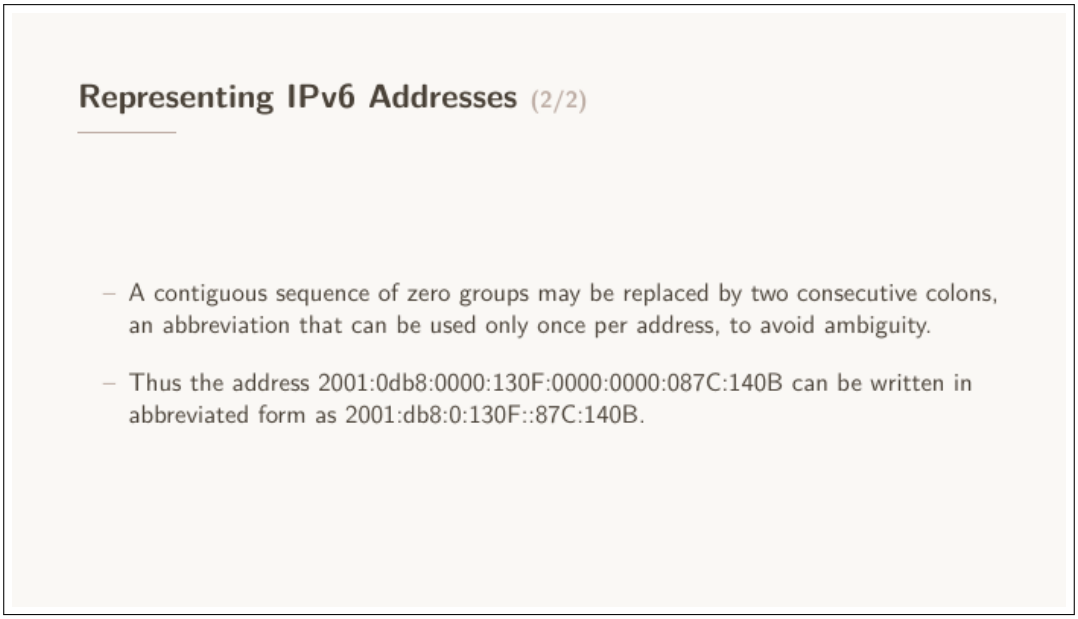
An unprecedented order of magnitude. The number of addresses is so large that addressing stops being a design constraint. For IoT, the result is that every sensor can have its own globally routable address, without translation.

Slide 24. Representing IPv6 Addresses (1/2)



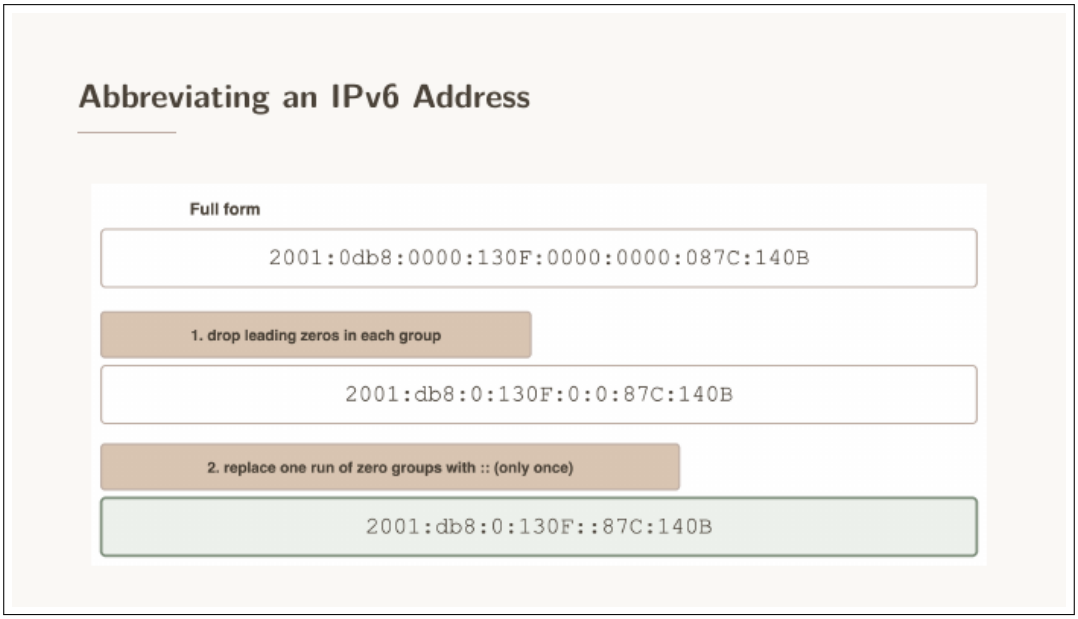
Writing an IPv6 address. Eight 16-bit groups in hexadecimal, separated by colons. Leading zeros in each group may be omitted. These rules exist because the full form is long and hard to transcribe without errors.

Slide 25. Representing IPv6 Addresses (2/2)



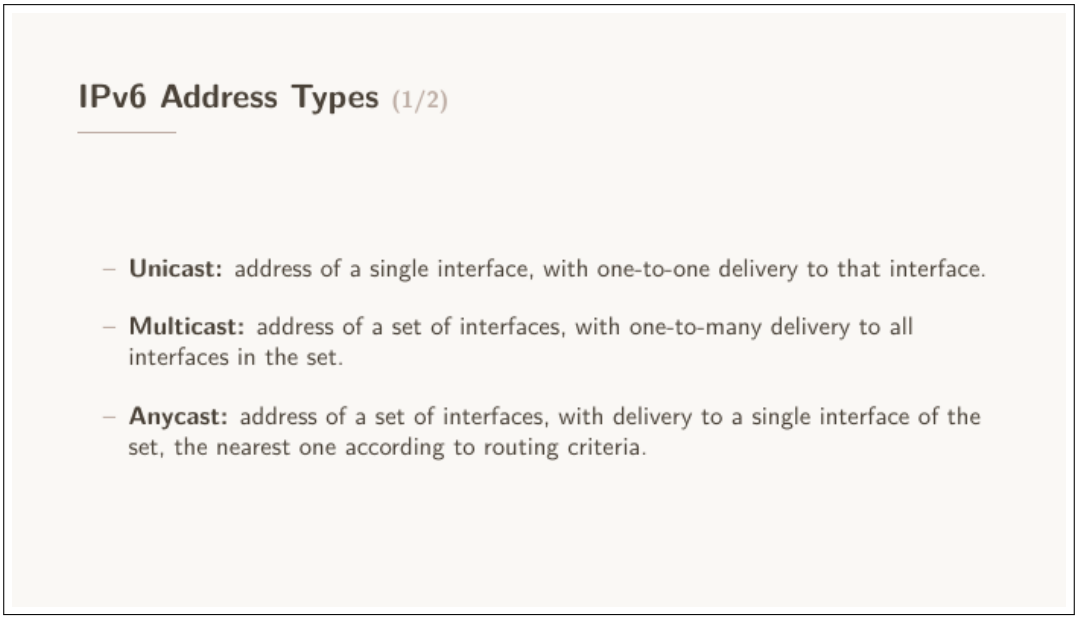
The double-colon abbreviation. A sequence of zero groups can be replaced by ::, but only once per address. The reason is ambiguity: with two :: it would be impossible to know how many zero groups each one stands for.

Slide 26. Abbreviating an IPv6 Address



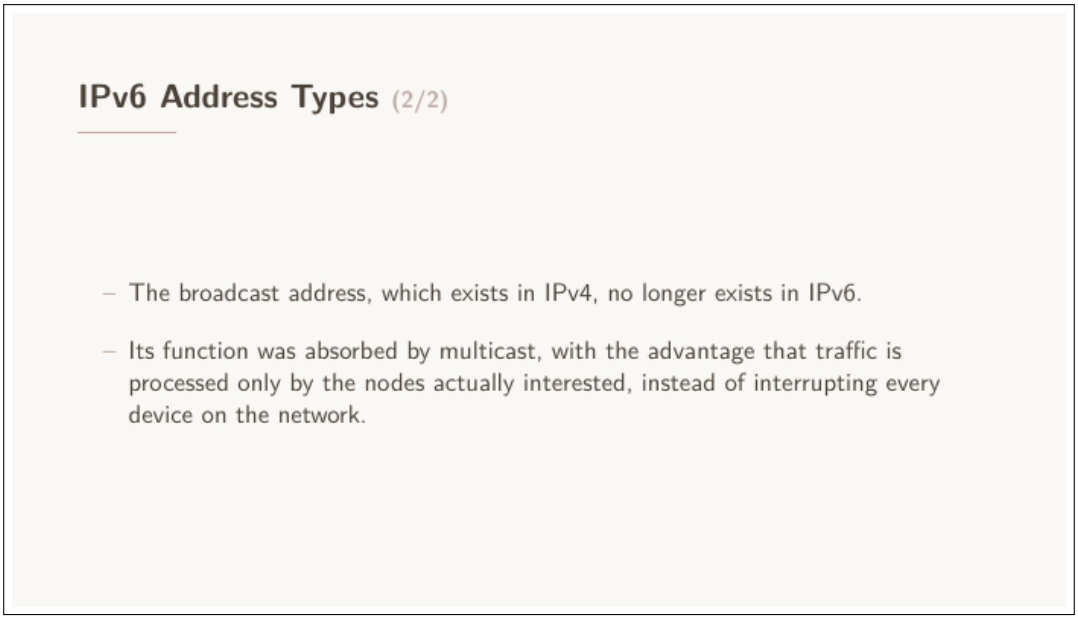
Applying the two rules. The figure applies the two rules to the same address, one at a time. Check: what would happen if 2001:db8:0:0:130F:0:0:87C were abbreviated with two ::? It would be ambiguous, because there are two sequences of zero groups.

Slide 27. IPv6 Address Types (1/2)



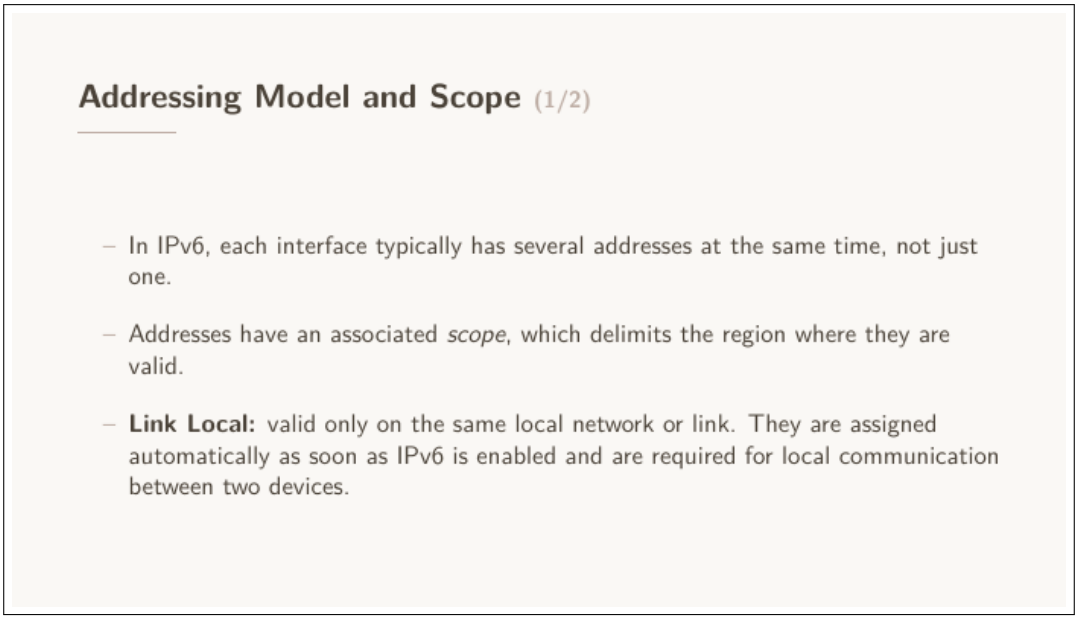
Three forms of delivery. Unicast delivers to one interface, multicast delivers to all those in a group and anycast delivers to the nearest one in a group. The difference between multicast and anycast is the one that confuses most: in the first, all members receive; in the second, only one.

Slide 28. IPv6 Address Types (2/2)



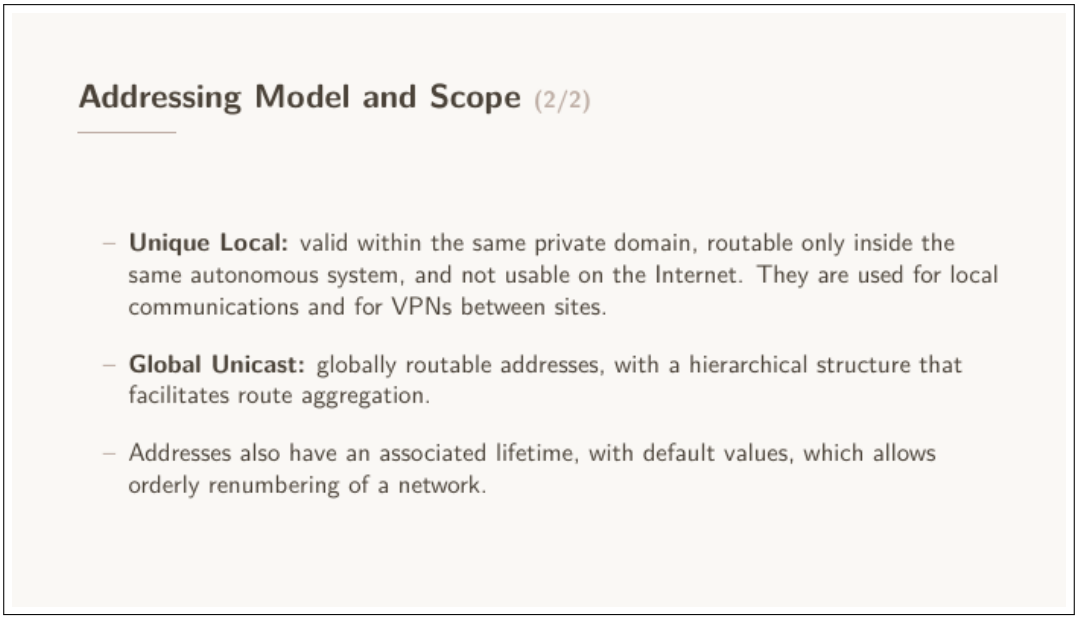
The end of broadcast. IPv4 broadcast forced every node to process every message, even those that did not concern it. IPv6 multicast does the same useful work without interrupting those who are not interested, which matters for devices that sleep to save energy.

Slide 29. Addressing Model and Scope (1/2)



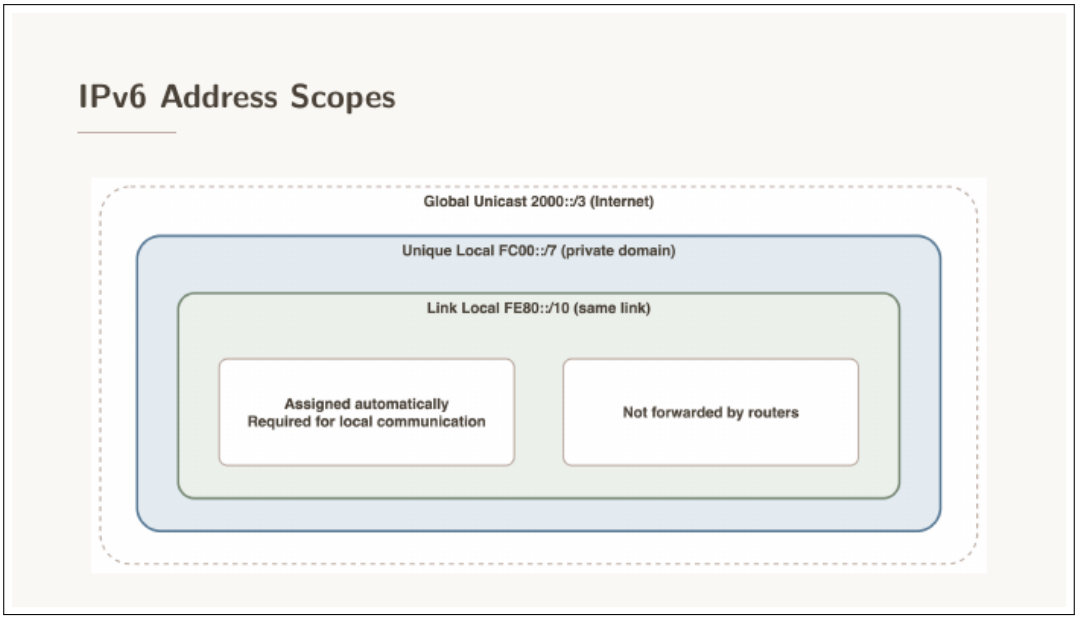
Several addresses per interface. An IPv6 interface normally has a link-local address and one or more global addresses at the same time. The scope indicates how far the address is valid. The link-local address always exists and is essential for the local operation of the protocol.

Slide 30. Addressing Model and Scope (2/2)



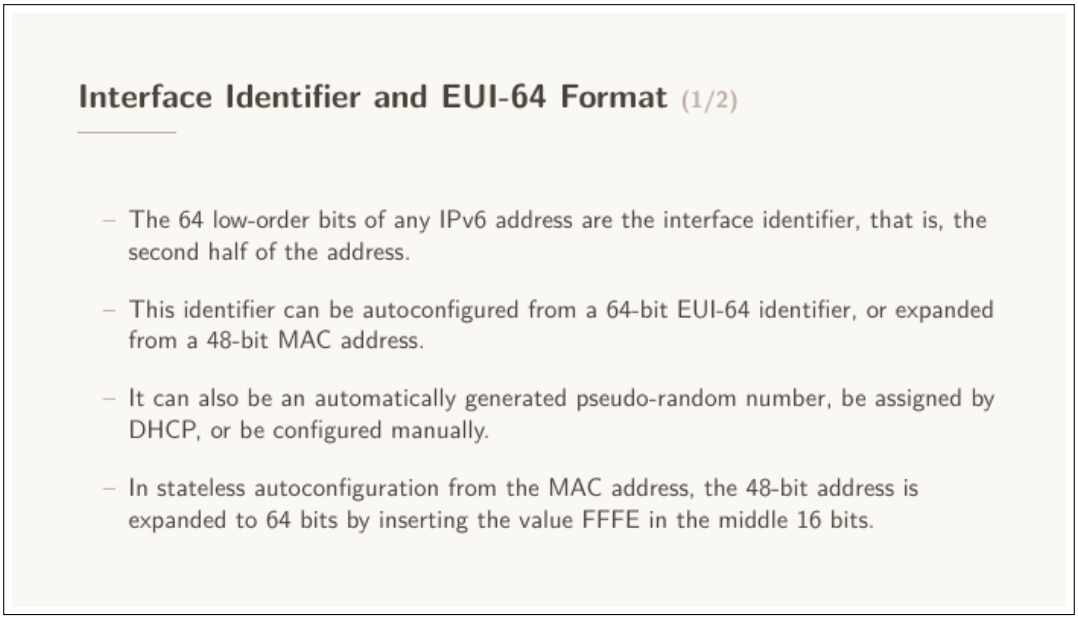
Unique local and global. Unique local addresses serve private communications within a domain, without leaving for the Internet, and play the role that private addresses play in IPv4. Global addresses are routable across the whole Internet and have a hierarchical structure, which allows routes to be aggregated.

Slide 31. IPv6 Address Scopes



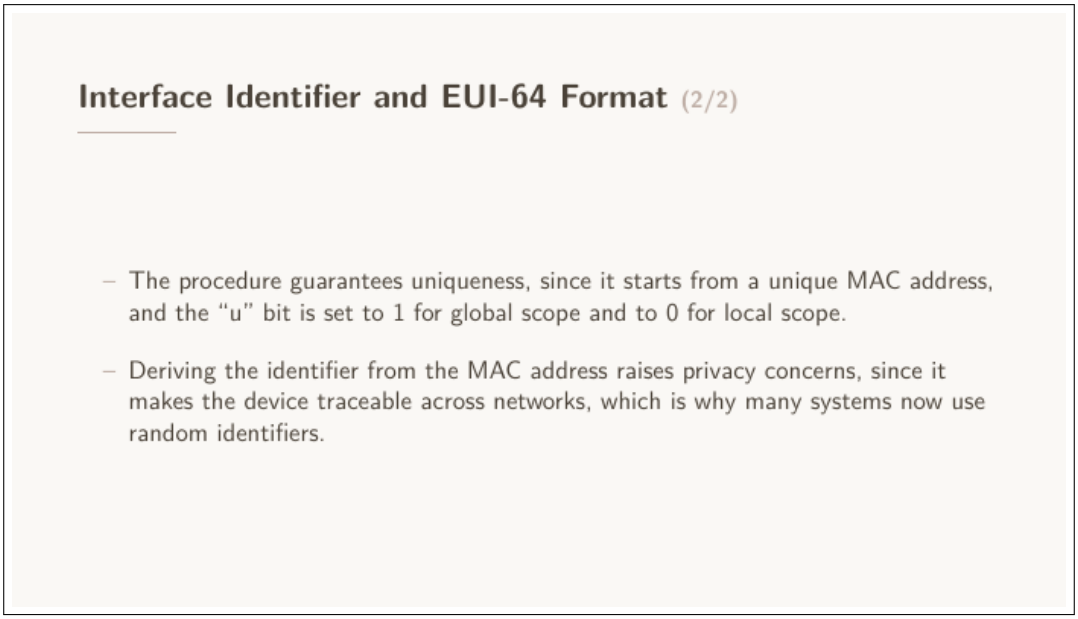
Concentric scopes. The three scopes are contained in one another: link-local is the smallest, unique local contains it, and global is the widest. A link-local address never crosses a router.

Slide 32. Interface Identifier and EUI-64 Format (1/2)



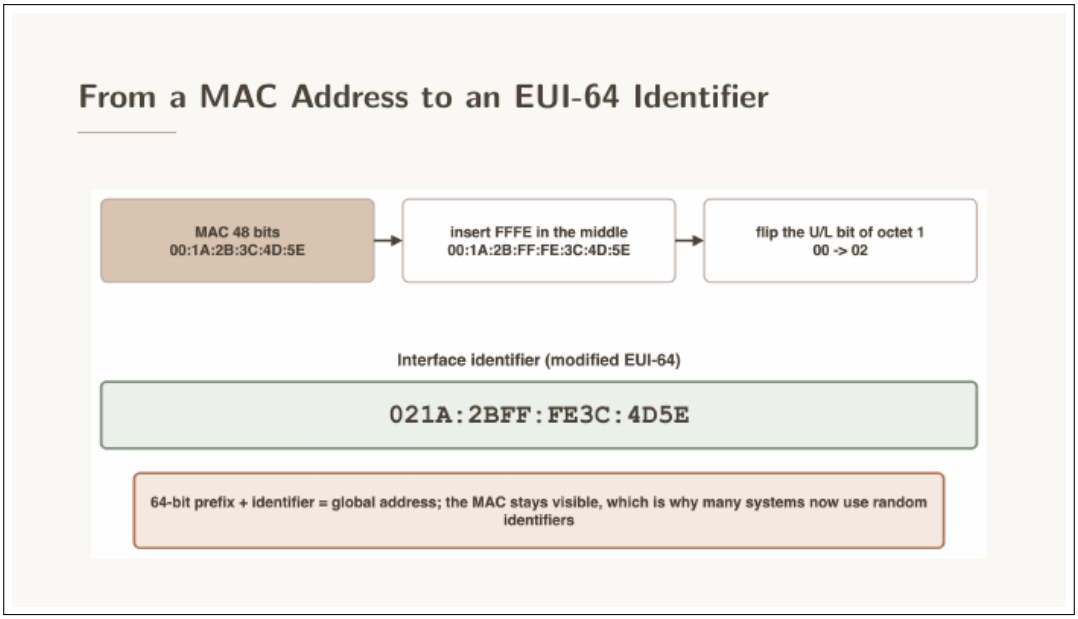
The second half of the address. The final 64 bits identify the interface within the network. They can be derived from the MAC address, be random, be assigned by DHCP or be configured by hand. The classic mechanism inserts FFFE in the middle of the MAC to obtain 64 bits.

Slide 33. Interface Identifier and EUI-64 Format (2/2)



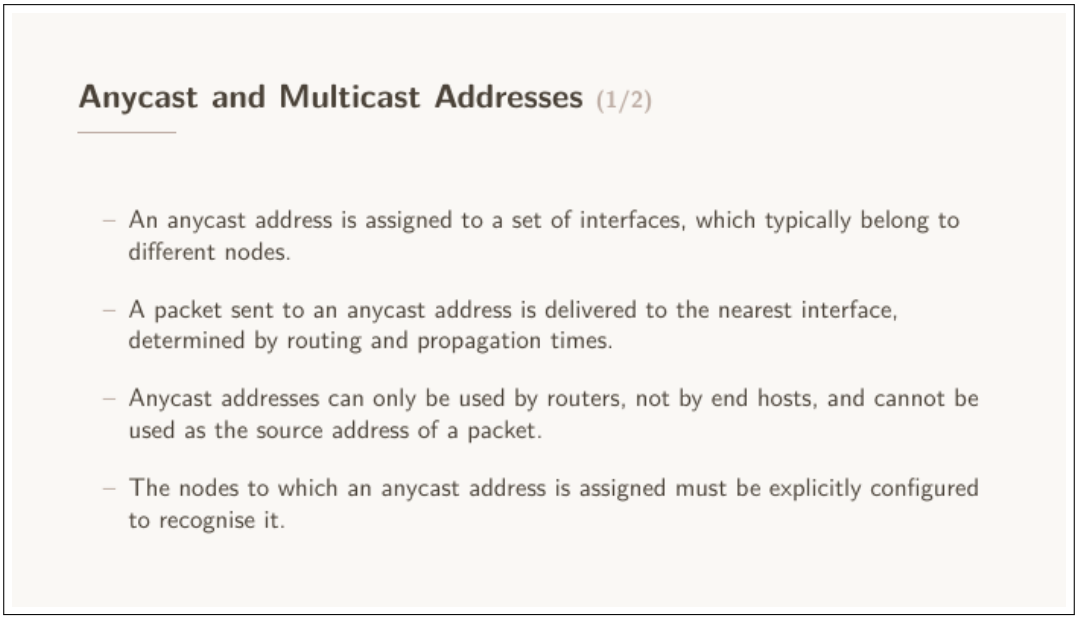
Uniqueness and privacy. Starting from the MAC guarantees uniqueness, but makes the device identifiable on any network where it appears. That is why many systems now use stable or temporary random identifiers, at the cost of losing predictability.

Slide 34. From a MAC Address to an EUI-64 Identifier



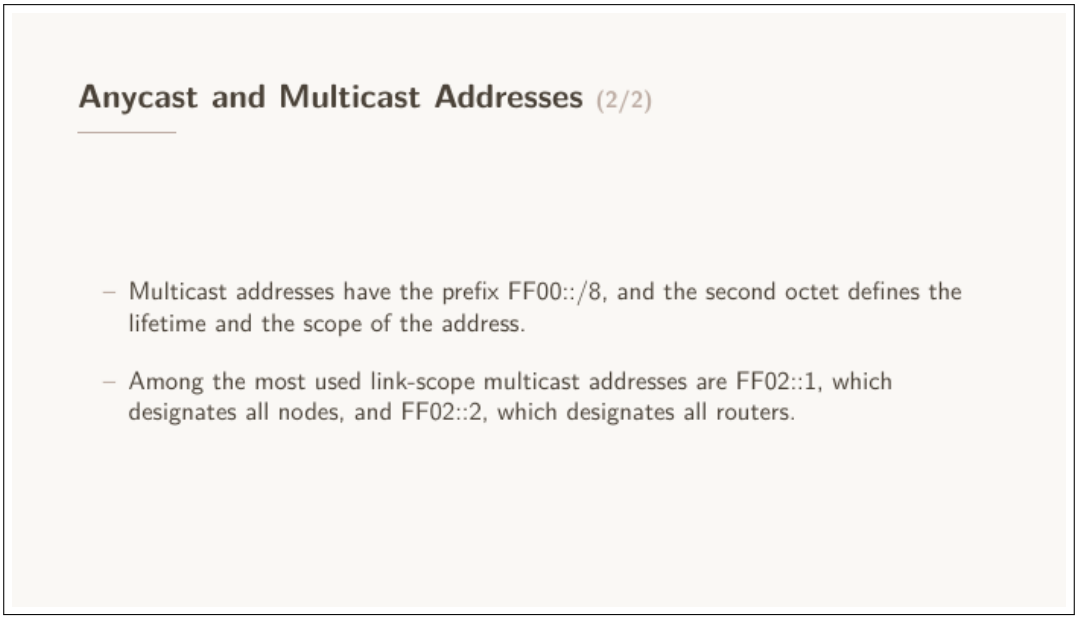
Computing an EUI-64 identifier. The three steps are in the figure: the 48-bit MAC is split in half, FFFE is inserted and the universal/local bit of the first octet is flipped. Applied to the MAC 00:1A:2B:3C:4D:5E, the result is 021A:2BFF:FE3C:4D5E. Check: can you explain why 00 becomes 02?

Slide 35. Anycast and Multicast Addresses (1/2)



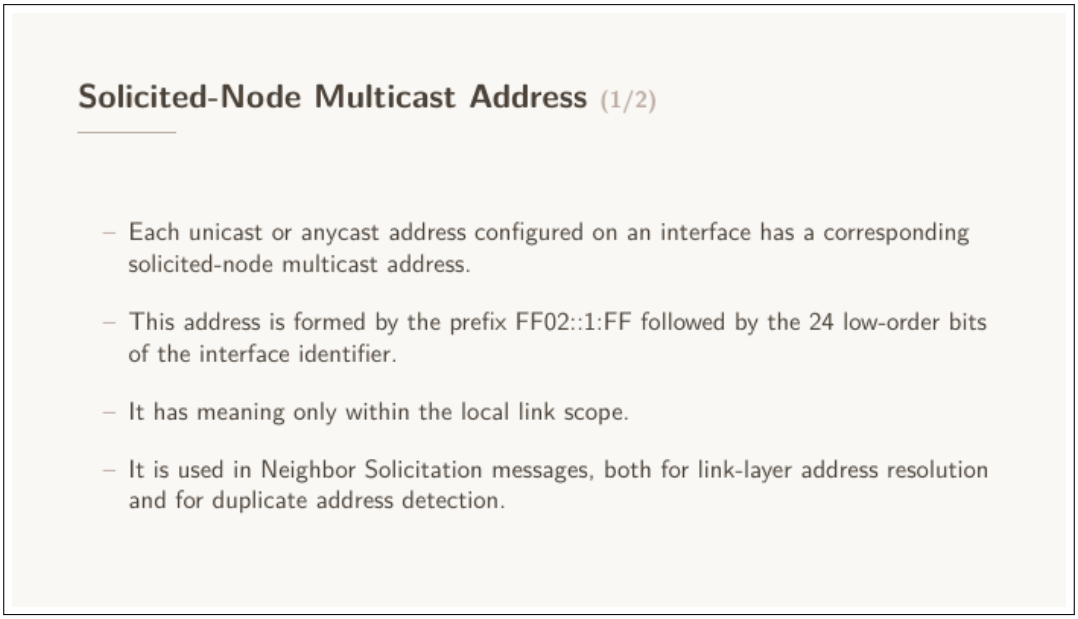
Anycast: the nearest one. An anycast address is shared by several nodes, and routing delivers to the nearest one. It is useful for replicated services such as DNS servers. Only routers use it as a destination, and never as a source.

Slide 36. Anycast and Multicast Addresses (2/2)



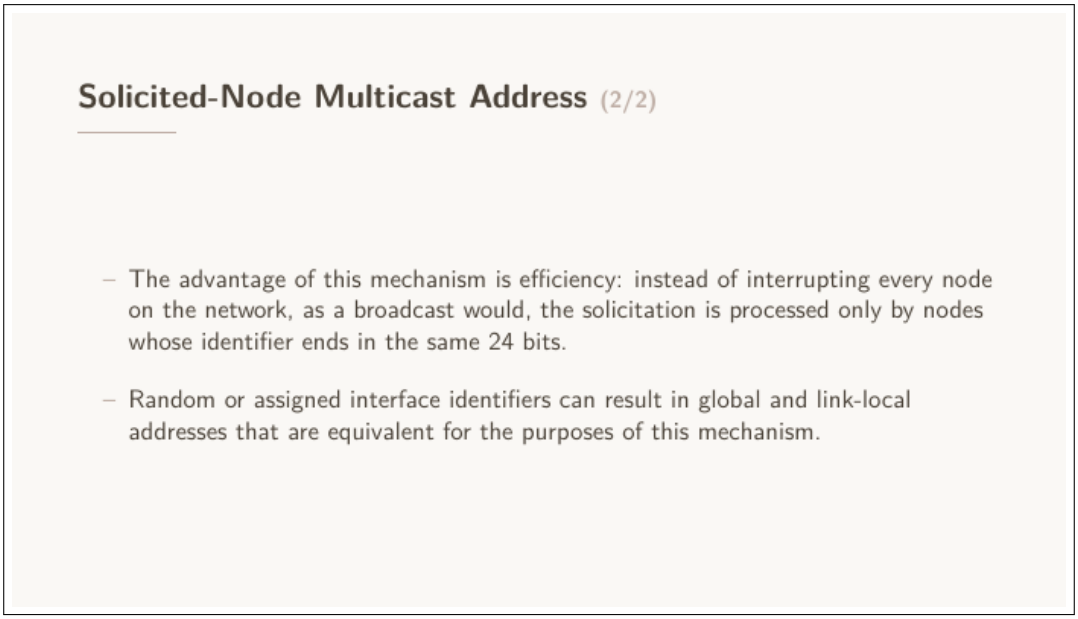
The well-known multicast addresses. The prefix FF00::/8 identifies multicast. FF02::1 reaches all nodes on the link and FF02::2 all routers. These two addresses reappear whenever discovery traffic is analysed.

Slide 37. Solicited-Node Multicast Address (1/2)



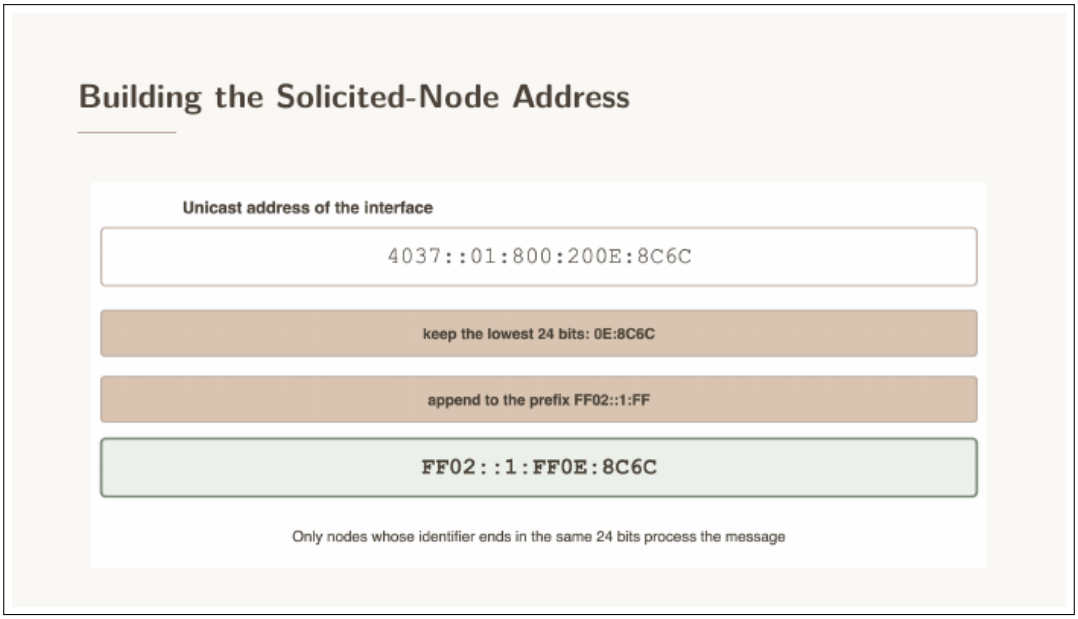
One group per address. Each unicast or anycast address of an interface has an associated solicited-node multicast address. It is a very small group, typically with one or a few members, which makes it efficient.

Slide 38. Solicited-Node Multicast Address (2/2)



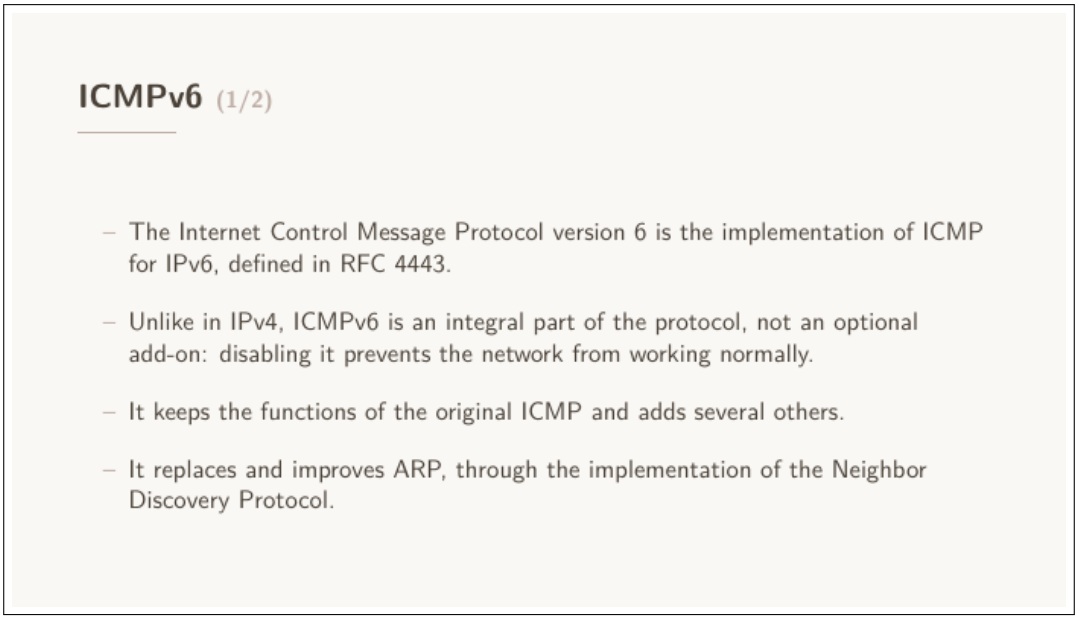
Efficiency instead of broadcast. Instead of asking the whole network, whoever is looking for a neighbour asks the group that only includes the nodes whose identifier ends in the same 24 bits. In practice, almost only the target receives the message.

Slide 39. Building the Solicited-Node Address



Building the solicited-node address. Take the last 24 bits of the unicast address and append them to the prefix FF02::1:FF. With the address 4037::01:800:200E:8C6C the result is FF02::1:FF0E:8C6C. Check: what would the solicited-node address of an address ending in ABCD:EF01 be?

Slide 40. ICMPv6 (1/2)



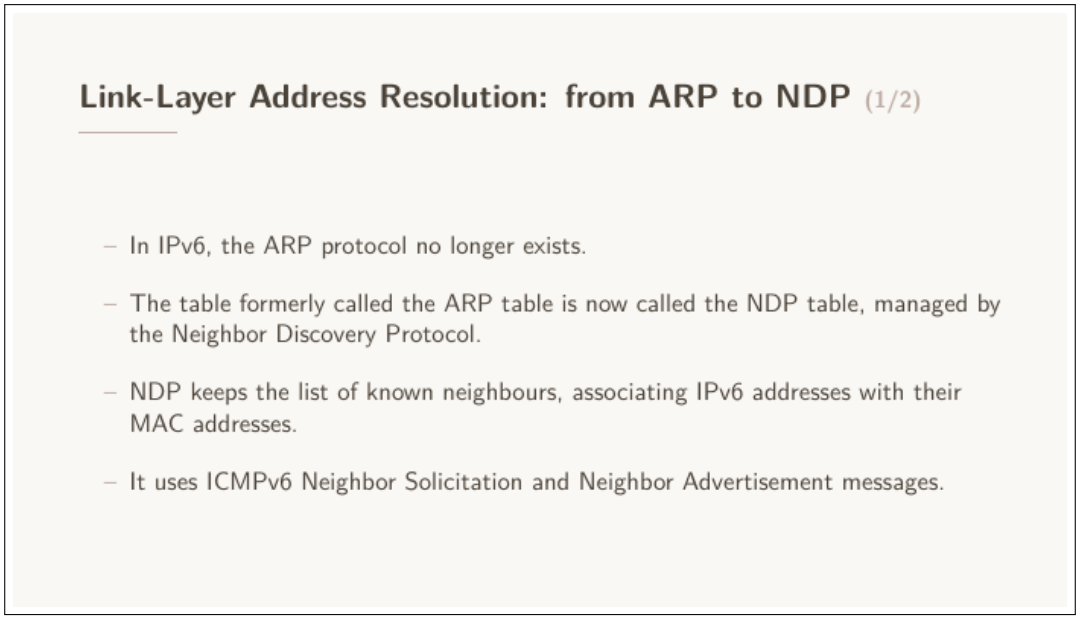
An indispensable protocol. In IPv4, ICMP was useful but dispensable. In IPv6, ICMPv6 is part of how the network works. Blocking ICMPv6 on a firewall, as was habitually done with ICMP, prevents neighbour discovery and autoconfiguration.

Slide 41. ICMPv6 (2/2)



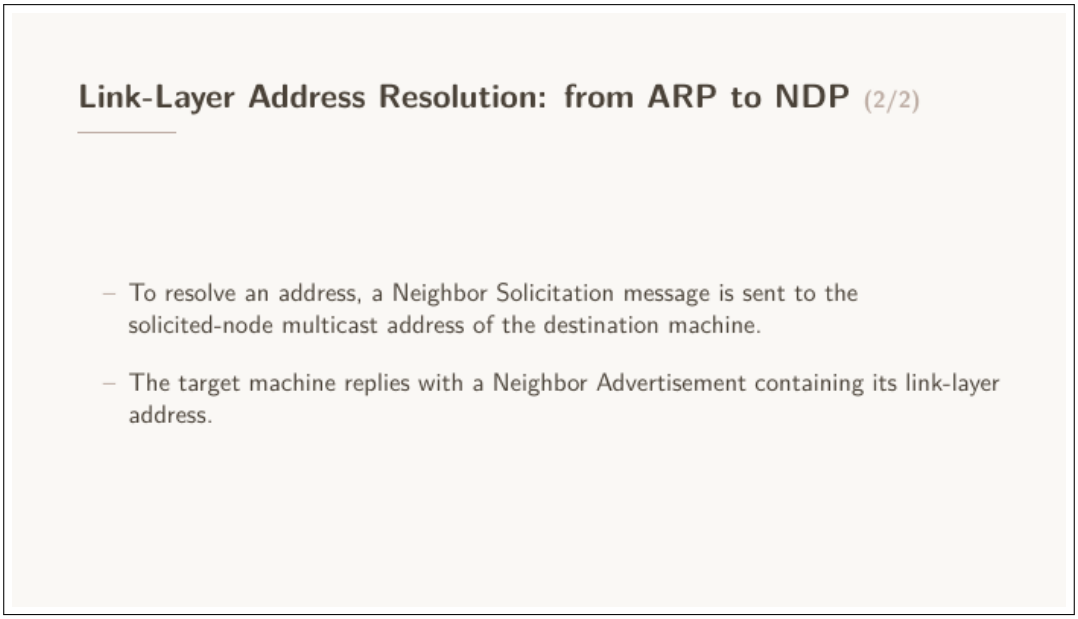
ICMPv6 functions. Router discovery, autoconfiguration, duplicate address detection and multicast group management. What in IPv4 was spread over several protocols is done in IPv6 with ICMPv6 messages.

Slide 42. Link-Layer Address Resolution: from ARP to NDP (1/2)



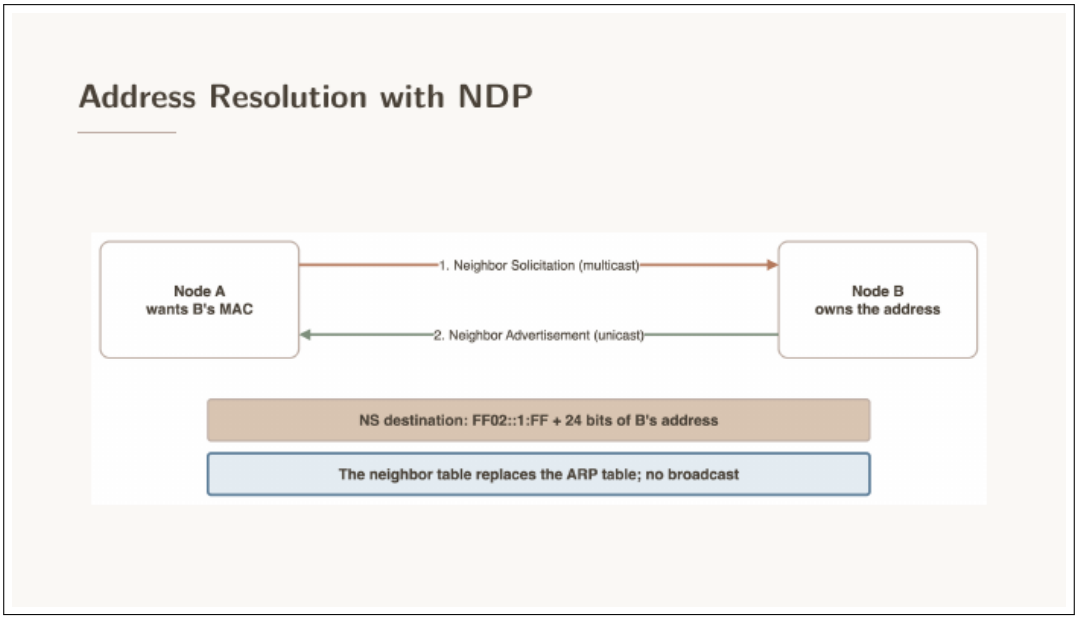
From ARP to NDP. ARP does not exist in IPv6. The association between IP address and physical address is done with the Neighbor Discovery Protocol, in ICMPv6 messages. The resulting table is the neighbour table.

Slide 43. Link-Layer Address Resolution: from ARP to NDP (2/2)



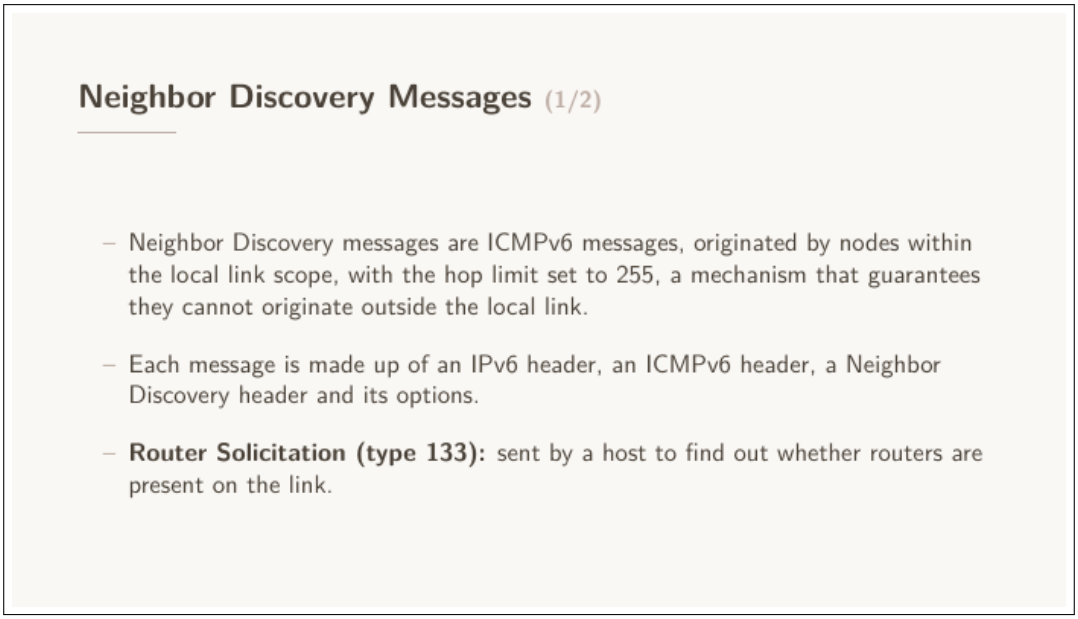
Resolving without broadcast. The solicitation goes to the target's solicited-node multicast address, and the target replies directly. ARP broadcast the request to the whole local network; NDP sends it only to whoever can own the address.

Slide 44. Address Resolution with NDP



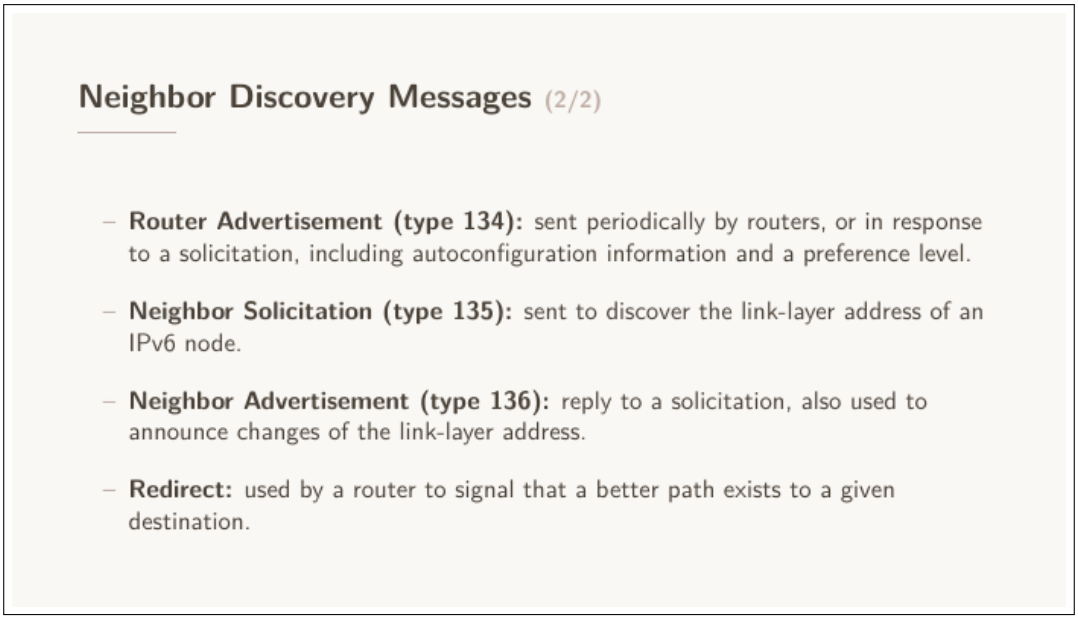
The message exchange. Two messages are enough: the Neighbor Solicitation, in multicast, and the Neighbor Advertisement, in unicast. The figure highlights the difference from ARP: there is no broadcast, and the reply carries the physical address node A needs.

Slide 45. Neighbor Discovery Messages (1/2)



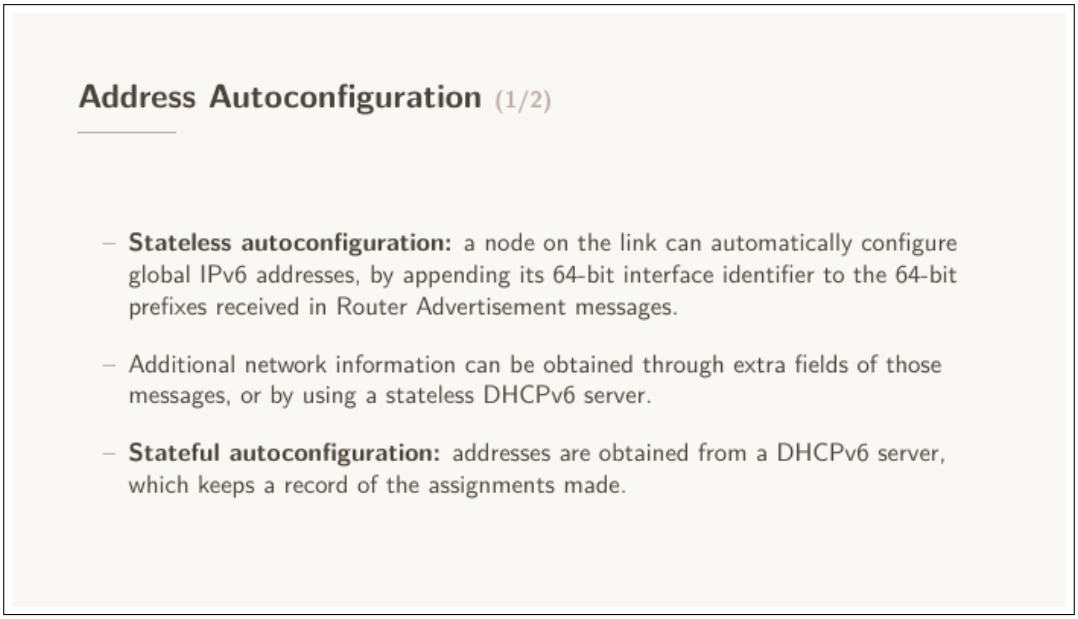
Discovery messages. All are ICMPv6 messages with a hop limit of 255. If the received value is different, the message passed through a router and must be ignored. It is a simple mechanism that guarantees only nodes on the same link take part.

Slide 46. Neighbor Discovery Messages (2/2)



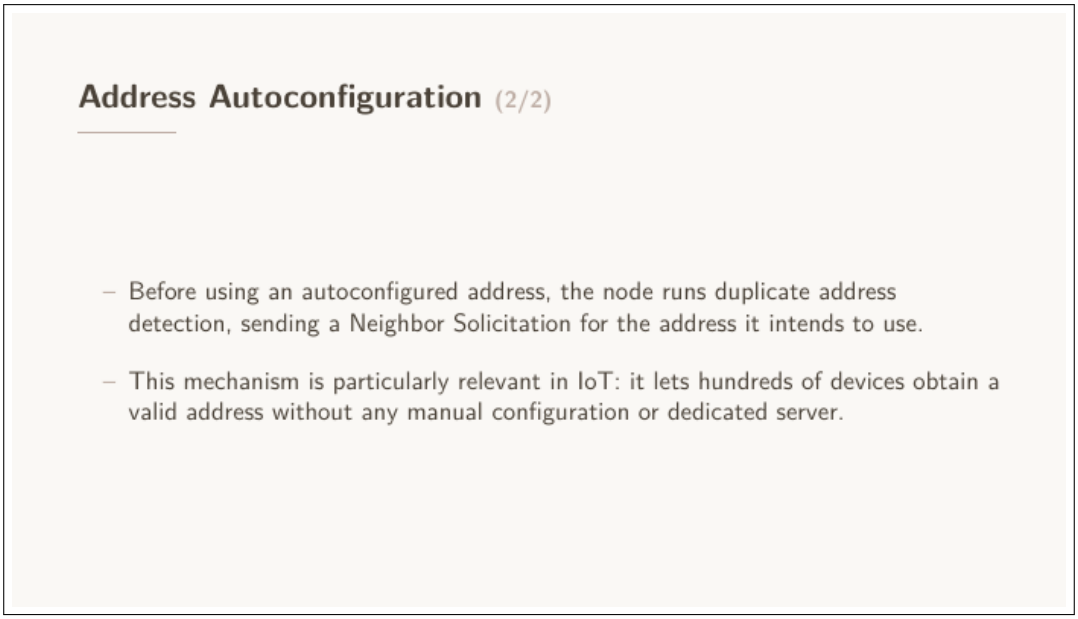
The remaining four messages. Types 134, 135 and 136 and the Redirect message complete the set. You should be able to associate each message with its function: 133 and 134 deal with routers, 135 and 136 with neighbours.

Slide 47. Address Autoconfiguration (1/2)



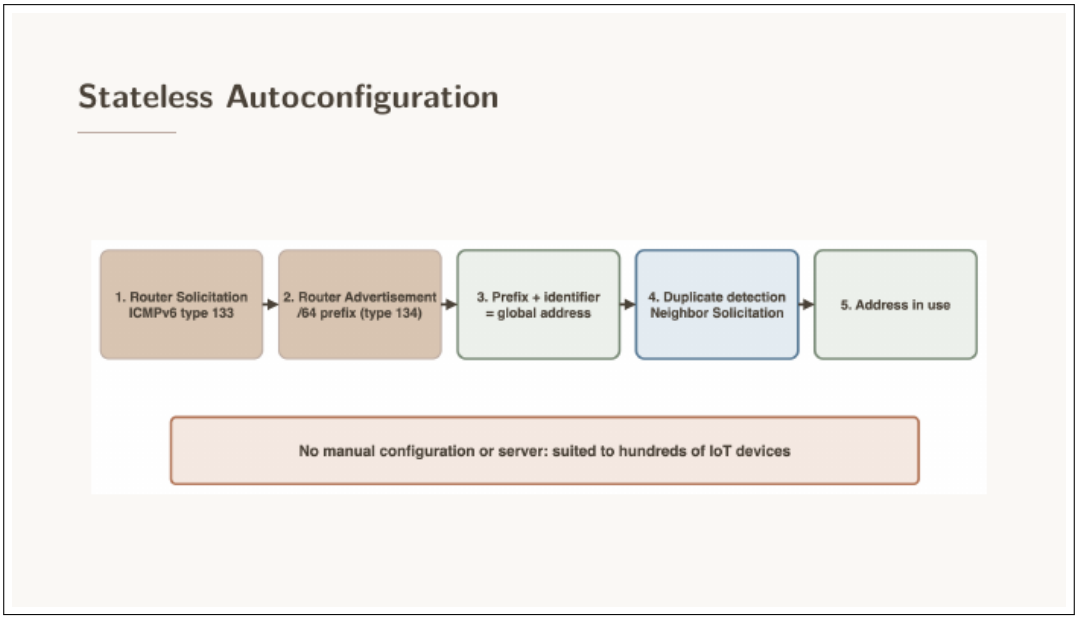
Two forms of autoconfiguration. Stateless, the node builds the address on its own from the prefix announced by the router. Stateful, it asks a DHCPv6 server, which records the assignments. The choice depends on whether central control of assignments is wanted.

Slide 48. Address Autoconfiguration (2/2)



Check before use. Before adopting an address, the node asks whether anyone already uses it. This is duplicate address detection. In IoT, this set of steps is what allows hundreds of devices to be installed without manual intervention.

Slide 49. Stateless Autoconfiguration



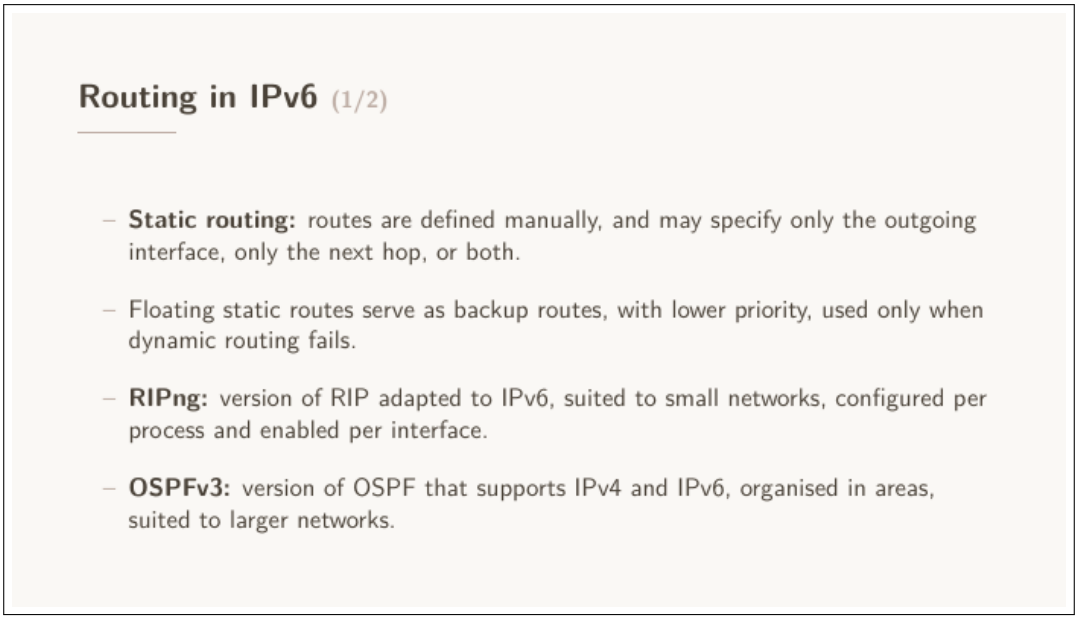
The stateless autoconfiguration sequence. The figure shows the five steps: the node asks, the router announces, the node composes prefix and identifier, checks for duplicates and starts using the address. No server keeps state, which reduces points of failure.

Slide 50. Multicast Listener Discovery



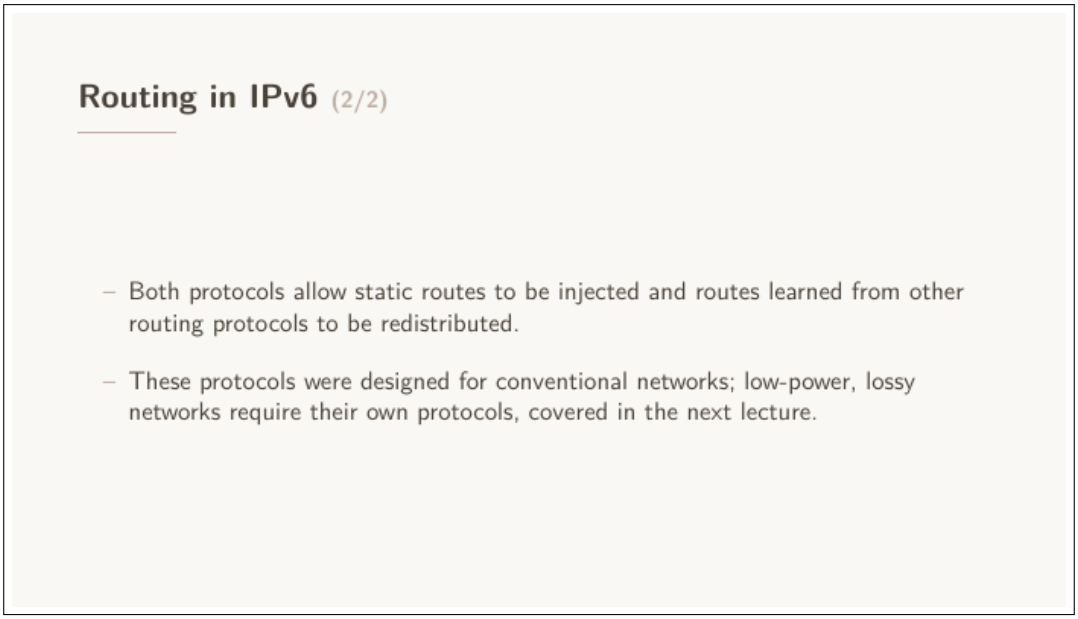
Multicast Listener Discovery. MLD lets nodes declare which multicast groups they are interested in, and lets routers know which links to forward that traffic to. Without it, multicast would be handled like broadcast.

Slide 51. Routing in IPv6 (1/2)



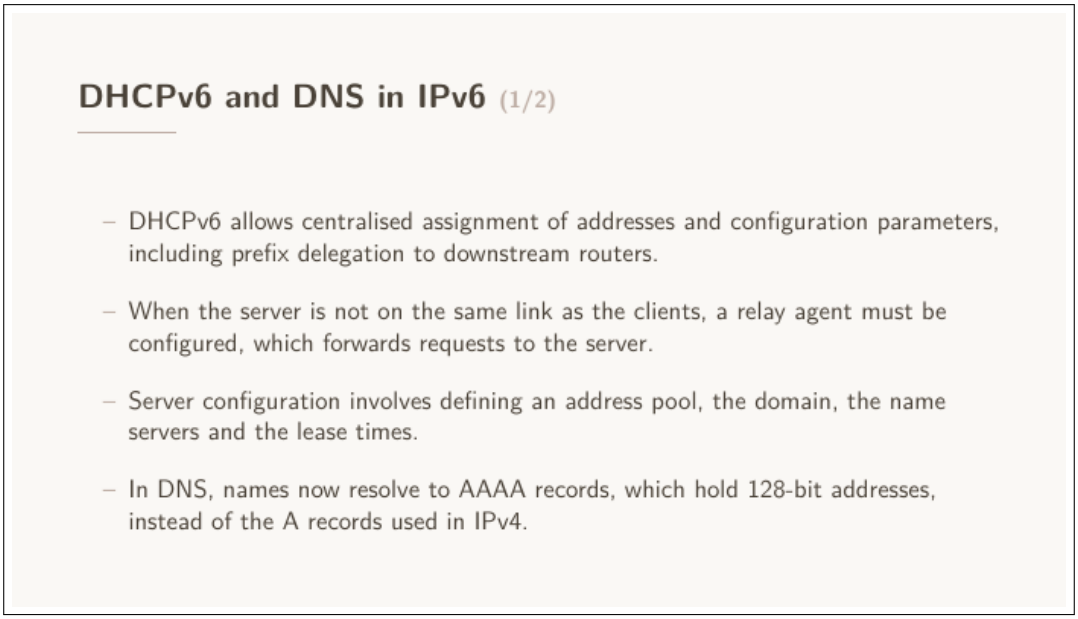
Routing protocols. Static routes serve small, stable networks. RIPng and OSPFv3 adapt well-known protocols to IPv6. The floating static route is a useful technique: it stays in reserve with lower priority and only comes into play when the dynamic route fails.

Slide 52. Routing in IPv6 (2/2)



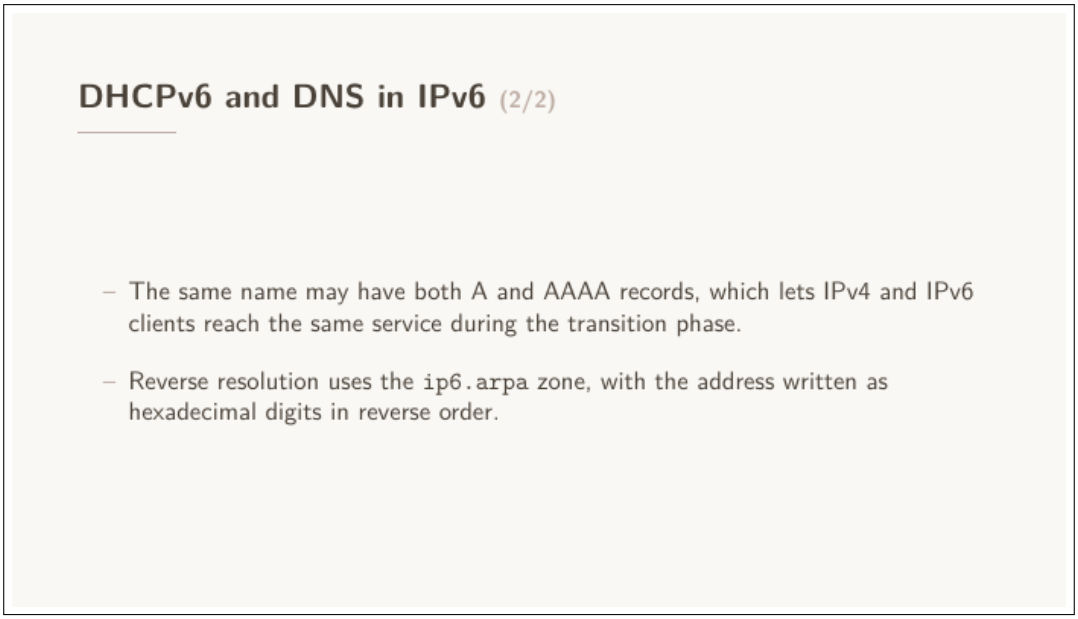
The limit of these protocols. RIPng and OSPFv3 were designed for conventional networks. Low-power, lossy networks, typical of IoT, require other protocols. This is the bridge to the next lectures.

Slide 53. DHCPv6 and DNS in IPv6 (1/2)



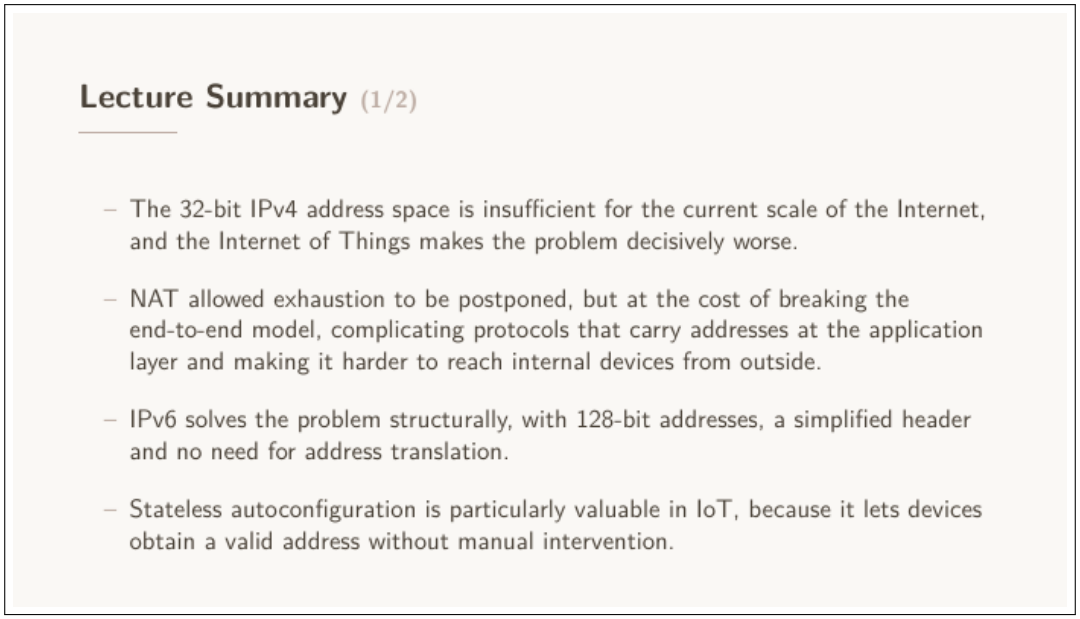
Central assignment and names. DHCPv6 centralises address assignment and prefix delegation. In DNS, AAAA records hold 128-bit addresses. A relay agent is needed when the server is not on the same link as the clients.

Slide 54. DHCPv6 and DNS in IPv6 (2/2)



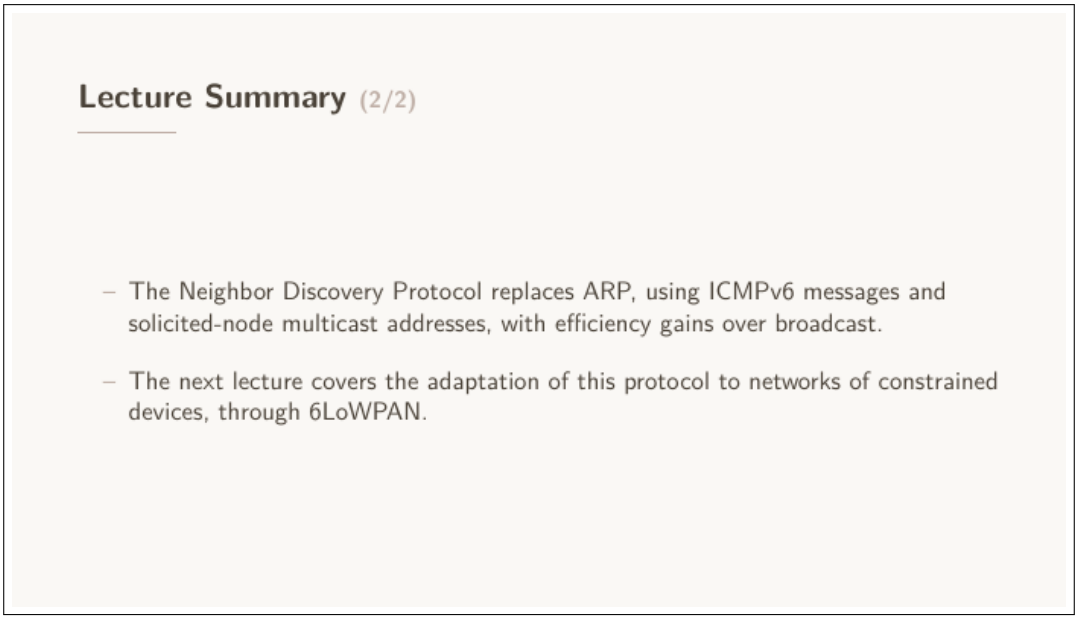
Coexistence during the transition. A name can have A and AAAA records at the same time, and the client chooses. Reverse resolution is done in the ip6.arpa zone, with the hexadecimal digits of the address written in reverse order.

Slide 55. Lecture Summary (1/2)



The essentials from IPv4 to IPv6. The 32-bit space ran out, NAT postponed the problem at the expense of the end-to-end model and IPv6 solves it structurally. Stateless autoconfiguration is the feature that matters most for IoT.

Slide 56. Lecture Summary (2/2)



What comes next. NDP replaces ARP with efficiency gains. The next lecture applies this knowledge to constrained devices, where full IPv6 is too heavy and headers have to be compressed.

Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. Why does NAT postpone the address exhaustion problem without solving it?
2. How is an EUI-64 identifier derived from the MAC address 00:1A:2B:3C:4D:5E?
3. What is the solicited-node multicast address for an interface ending in ...:200E:8C6C?
4. What is the difference between multicast and anycast delivery?
5. Which messages take part in stateless autoconfiguration, and in what order?
6. Why must ICMPv6 not be blocked on a firewall?

References

- S. Deering, R. Hinden. *Internet Protocol, Version 6 (IPv6) Specification.* RFC 8200, 2017.
- T. Narten et al. *Neighbor Discovery for IP version 6 (IPv6).* RFC 4861, 2007.
- S. Thomson, T. Narten, T. Jinmei. *IPv6 Stateless Address Autoconfiguration.* RFC 4862, 2007.
- A. Conta, S. Deering, M. Gupta. *ICMPv6 for IPv6.* RFC 4443, 2006.